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Frequency Filtering

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Lecture Roadmap

1. Images as sums of sinusoidal patterns
2. Sampling, spectral replicas, and aliasing
3. Reading and computing the 2D DFT
4. Magnitude, phase, and frequency-domain filtering

Filtering in the Frequency Domain

Spatial frequency: how rapidly image intensity changes across position.

A Fourier transform describes an image using sinusoidal components.

A frequency-domain filter changes selected components.

Guiding question: which frequencies should we keep?

Opening Question: Image Frequencies

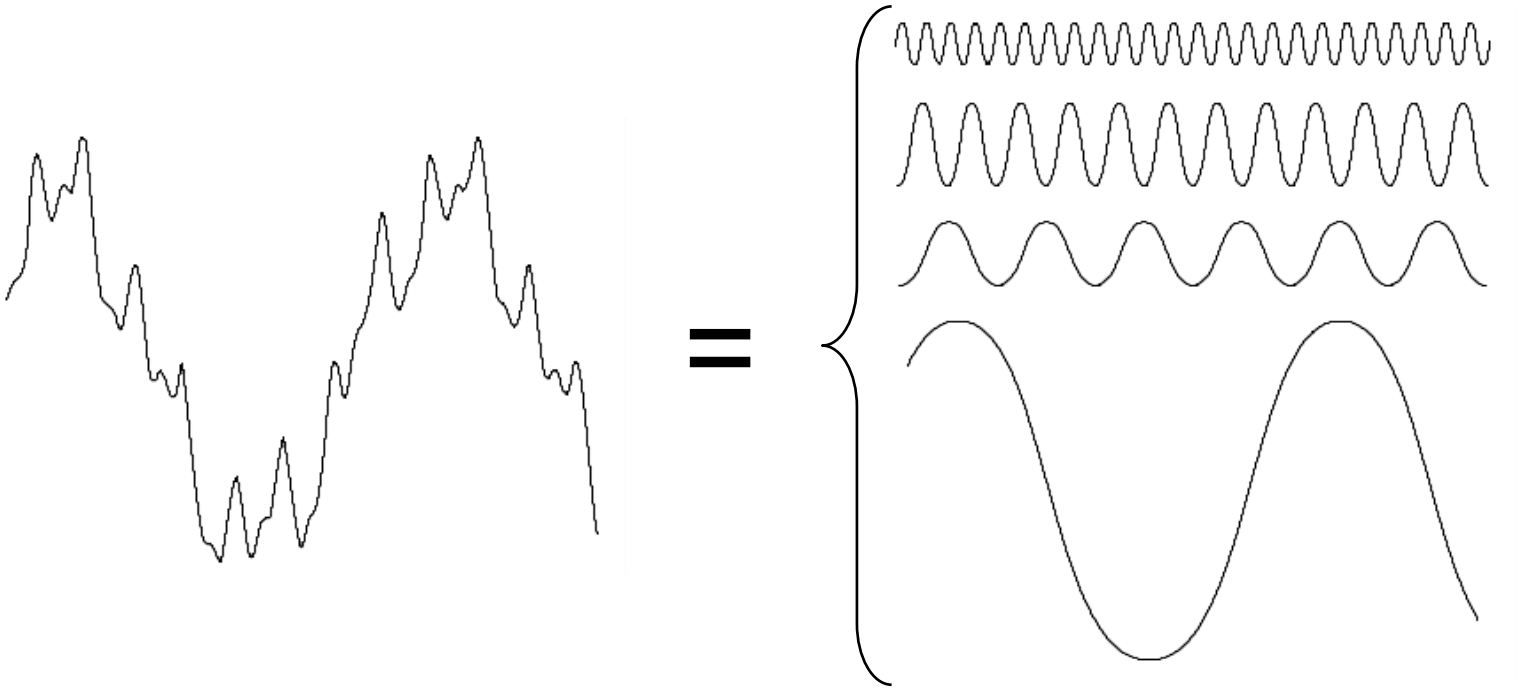
Imagine vertical stripes across an image.

If the stripes become closer together, what changes?

Where would their peaks appear in a centered spectrum?

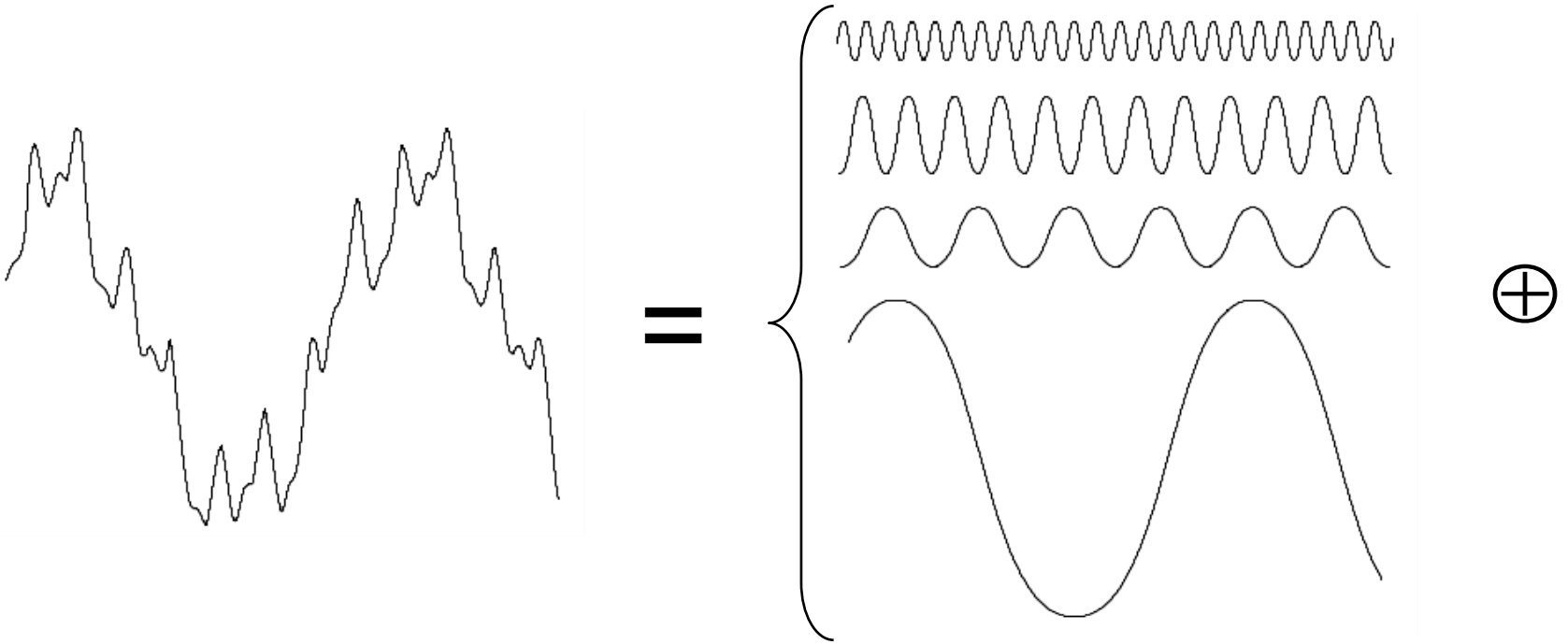
Think first. We will return to this after the 2D transform.

The Big Idea



Any function that periodically repeats itself can be expressed as a sum of sines and cosines of different frequencies each multiplied by a different coefficient – a *Fourier series*

The Big Idea

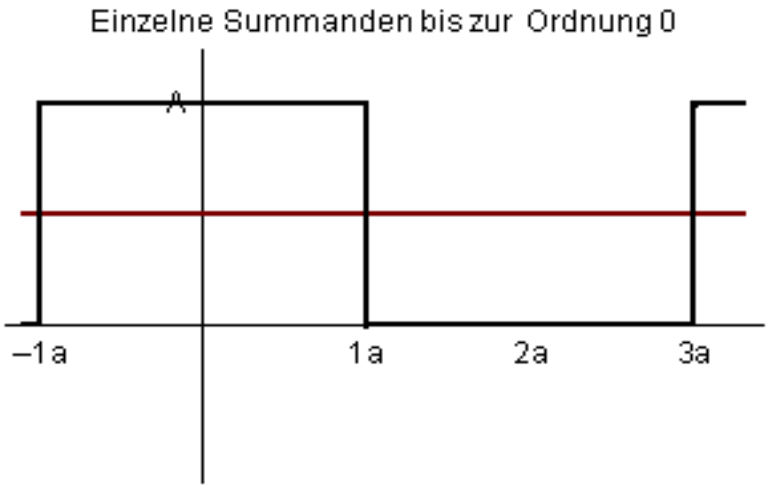


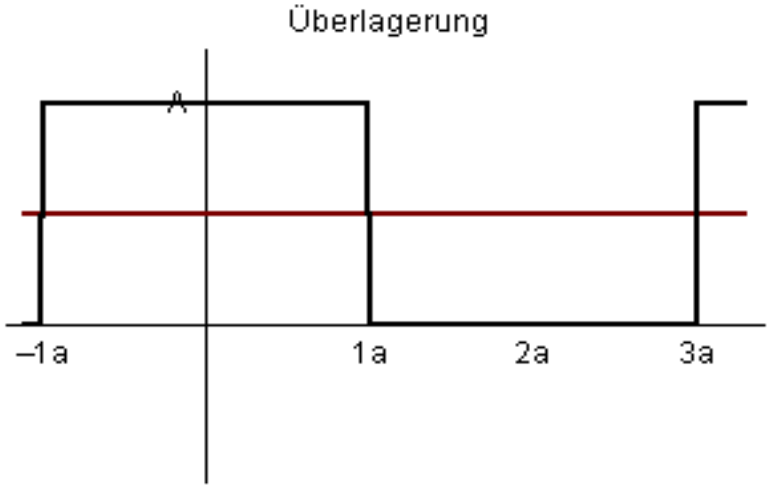
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{l} + b_n \sin\frac{n\pi x}{l} \right)$$

$$a_n = \int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}\,dx, \quad (n = 0,1,2,\ldots)$$

$$b_n = \int_{-l}^{l} f(x)\sin\frac{n\pi x}{l}\,dx, \quad (n = 1,2,3\ldots)$$

The Big Idea



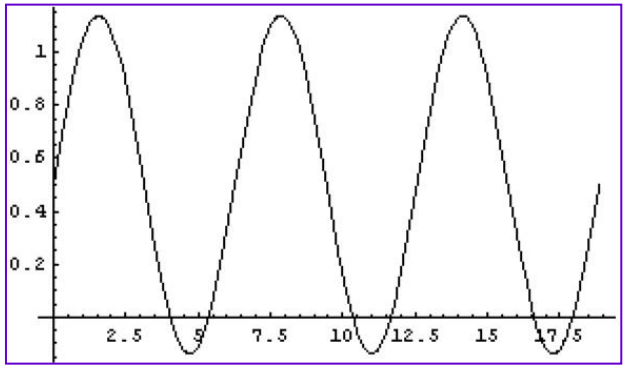


Notice how we get closer and closer to the original function as we add more and more frequencies

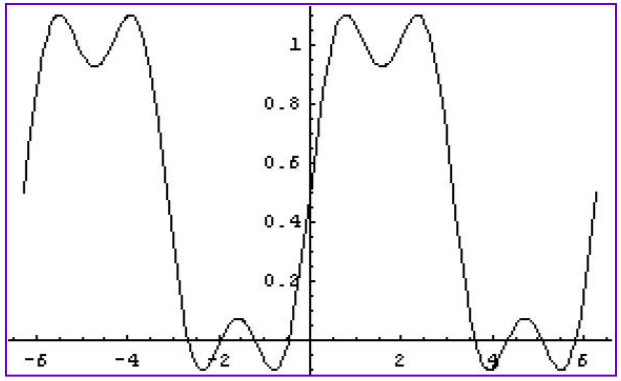
Fourier Series

- The Fourier series expansion of a periodic signal $f(t)$.

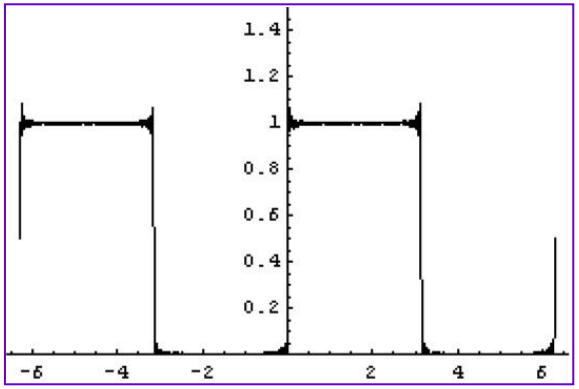
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{l} + b_n \sin\frac{n\pi x}{l} \right)$$



$$\frac{1}{2} + \frac{2}{\pi}\sin(x)$$



$$\frac{1}{2} + \frac{2}{\pi}\sin(x) + \frac{2}{3\pi}\sin(3x)$$



J.W. Gibbs phenomen

$$\frac{1}{2} + \frac{2}{\pi}\sum_{n=0}^{49}\sin\left((2n+1)x\right)$$

Note that the little spikes at the edge of the square wave are present even after including many hundreds of terms (although they become finer and finer). They are a consequence of trying to describe a discontinuous step function with smooth sine waves.

Check: Adding High Frequencies

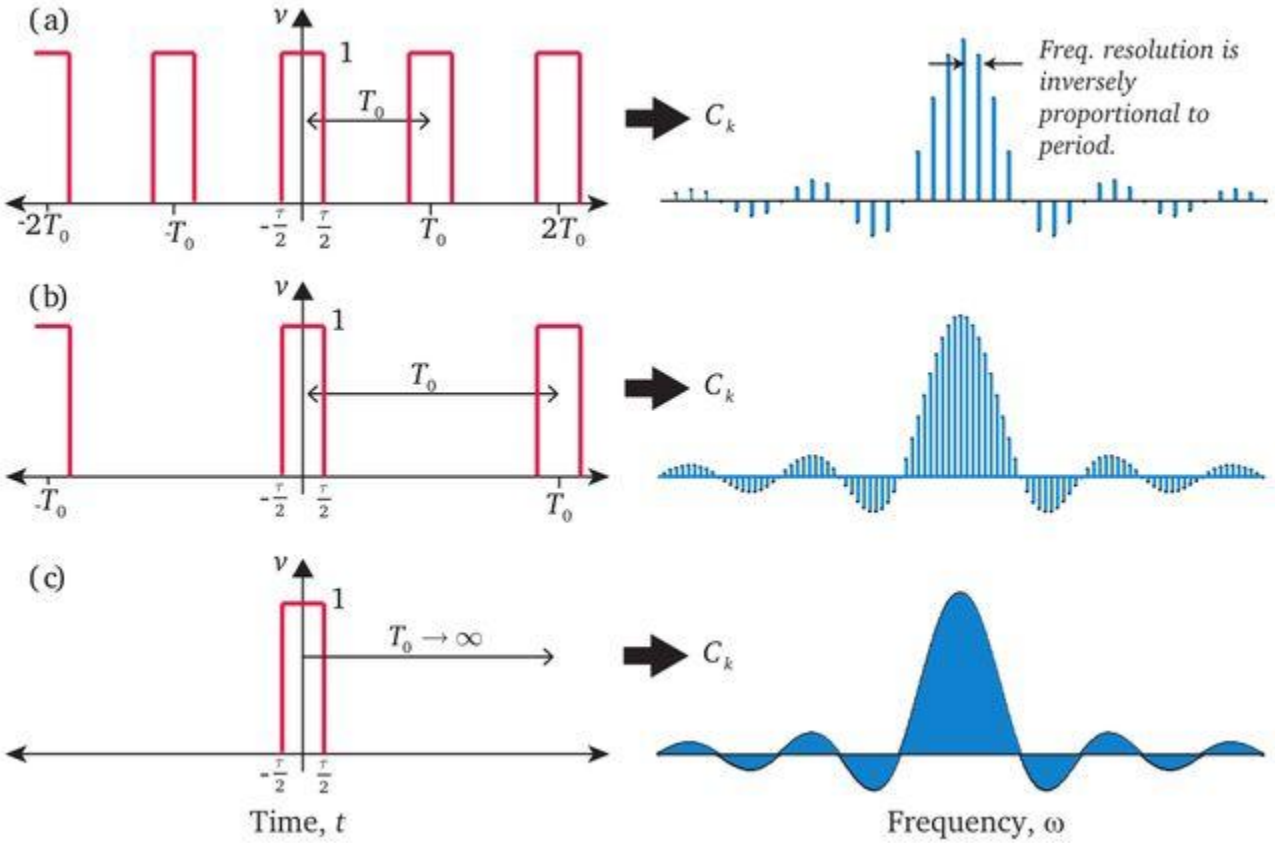
Which part of a square wave needs high frequencies?

A. Flat regions B. Sharp transitions

With more harmonics, does the edge overshoot vanish?

Predict before looking back at the reconstruction.

Fourier Series and Fourier Transform



1D continuous signals

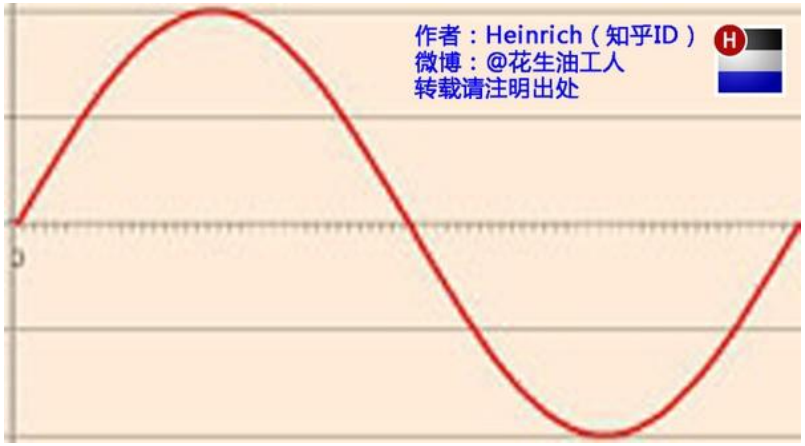
- The Fourier transform of a continuous signal $f(t)$.

$$F(\mu) = \int_{-\infty}^{+\infty} f(t) e^{-j2\pi\mu t} dt$$

$$f(t) = \int_{-\infty}^{+\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

- Attention: the variable is the frequency (Hz) and not the radial frequency ($\omega=2\pi\mu$) as in the Signals and Systems course.

“如果看了此文你还不懂傅里叶变换,那就过来掐死我吧”





http://www.jezzamon.com/fourier/zh-cn.html

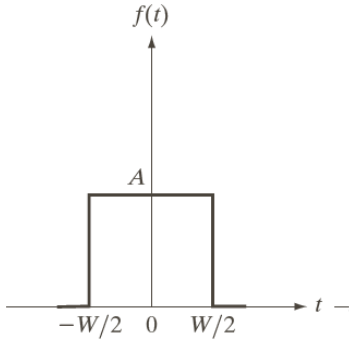
http://zhuanlan.zhihu.com/wille/19763358

1D continuous signals

Example: Obtaining the Fourier transform of a simple function

$$F(\mu) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi\mu t}\,\mathrm{dt} = \int_{-\frac{W}{2}}^{\frac{W}{2}} Ae^{-j2\pi\mu t}dt$$

$$= \frac{-A}{j2\pi\mu}\left[e^{-j2\pi\mu W/2} - e^{j2\pi\mu W/2}\right]$$



$$= \frac{A}{j2\pi\mu}\left[e^{j\pi\mu W} - e^{-j\pi\mu W}\right] \leftarrow \sin\theta = (e^{j\theta} - e^{-j\theta})/2j$$

$$= \frac{A}{j2\pi\mu} \cdot \sin(\pi\mu W) \cdot 2j$$

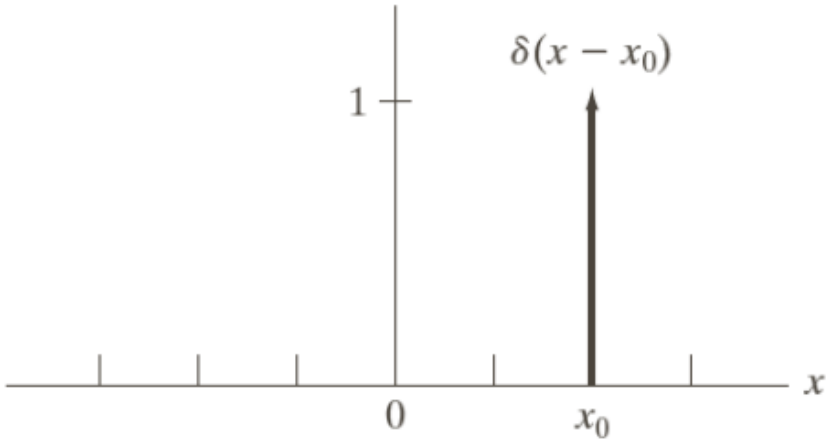
$$= AW\frac{sin(\pi\mu W)}{(\pi\mu W)} \leftarrow sinc(m) = \frac{\sin(\pi m)}{(\pi m)} = \begin{cases} 1, & m = 0 \\ 0, & m \neq 0 \end{cases}$$

$$= AW\ sinc(\pi\mu W)$$

***sinc* function: $sinc(0) = 1$, and $sinc(m) = 0$ for all other integer values of m**

Impulses

- It may be considered both as continuous and discrete.
- Useful for the representation of discrete signals through sampling of continuous signals.



Continuous

$$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(t)dt = 1$$

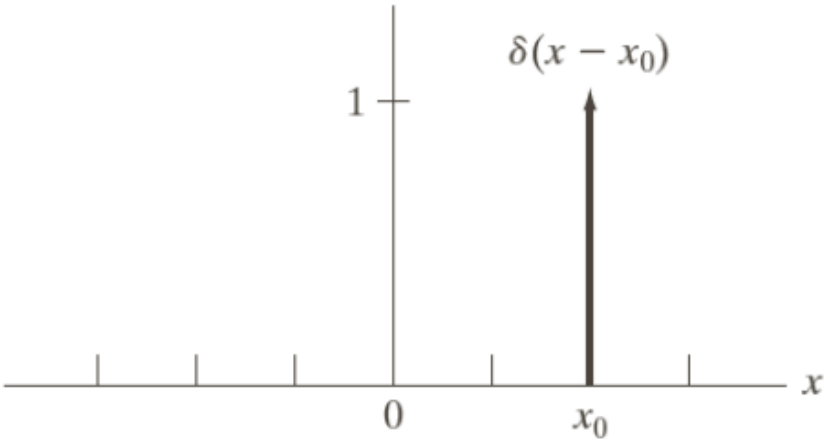
Discrete

$$\delta(x) = \begin{cases} 1, & x = 0 \\ 0, & x \neq 0 \end{cases}$$

$$\sum_{x=-\infty}^{\infty} \delta(x) = 1$$

Impulses

- It may be considered both as continuous and discrete.
- Useful for the representation of discrete signals through sampling of continuous signals.



$$\delta(x-x_0)=\begin{cases}1, & x=x_0\\ 0 & \text{otherwise}\end{cases}$$

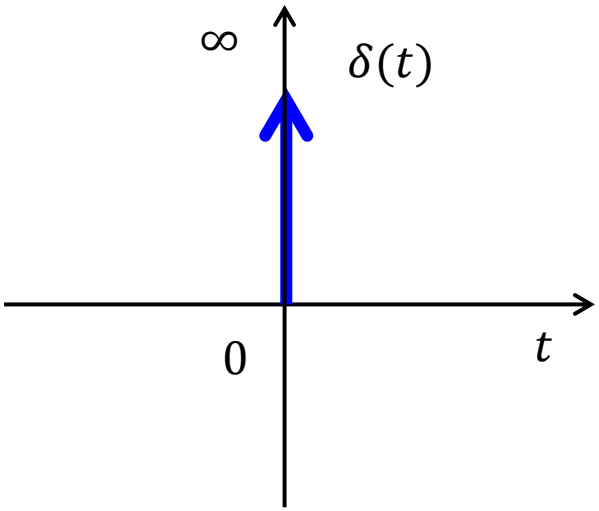
$$\int_{-\infty}^{+\infty} f(x)\delta(x-x_0)dx = f(x_0)$$

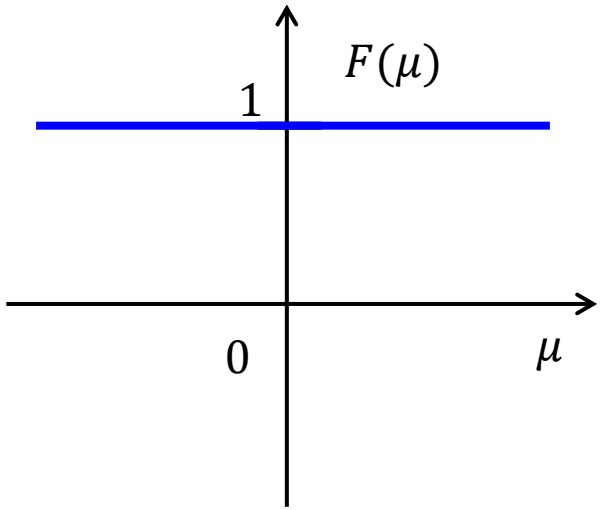
Impulses

- Intermediate result
  - The Fourier transform of an impulse

$$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$

$$F(\mu) = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi\mu t} dt = e^{-j2\pi\mu 0} = e^{0} = 1$$



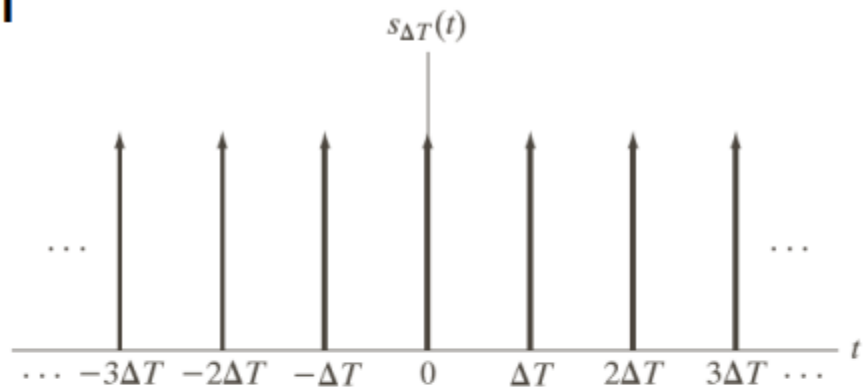


  - The FT of an impulse located at the origin of the spatial domain is a constant in frequency domain

Sampling

Impulse train function

$$S_{\Delta T}(t) = \sum_{n=-\infty}^{+\infty} \delta(t - n\Delta T)$$



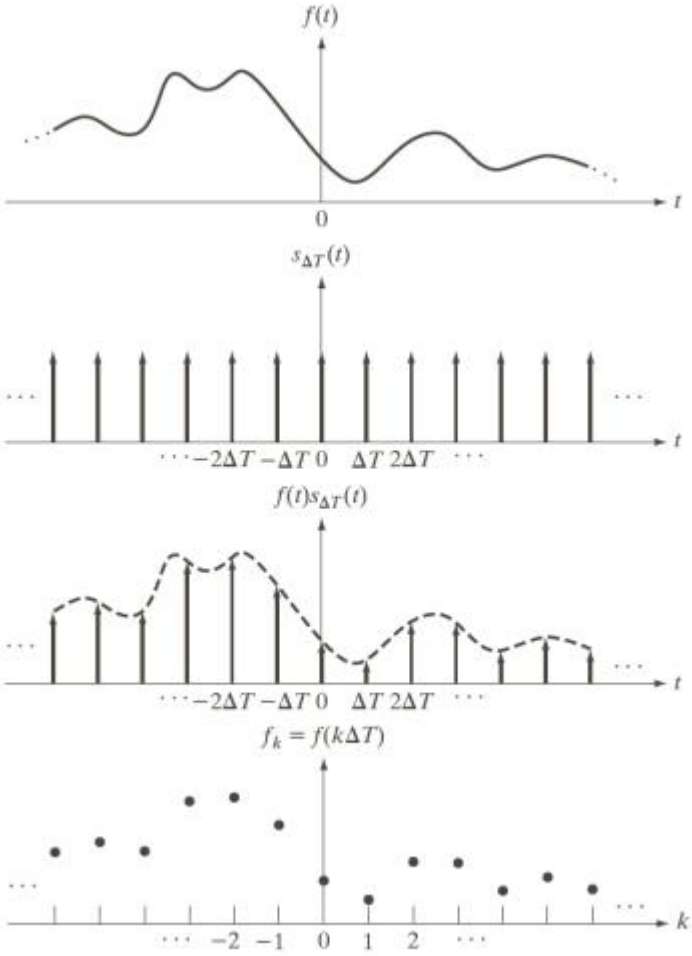
$$x[n] = x(t)S_{\Delta T}(t) = \sum_{n=-\infty}^{+\infty} x(t)\delta(t - n\Delta T) = \sum_{n=-\infty}^{+\infty} x(n\Delta T)\delta(t - n\Delta T)$$

Sampling

$$x[n] = x(t)S_{\Delta T}(t)$$

$$= \sum_{n=-\infty}^{+\infty} x(t)\delta(t - n\Delta T)$$

$$= \sum_{n=-\infty}^{+\infty} x(n\Delta T)\delta(t - n\Delta T)$$



Impulses

- Intermediate result
  - The Fourier transform of the impulse train

$$\sum_{n=-\infty}^{+\infty} \delta(t - n\Delta T) \leftrightarrow \frac{1}{\Delta T} \sum_{n=-\infty}^{+\infty} \delta\left(\mu - \frac{n}{\Delta T}\right)$$

- It is also an impulse train in the frequency domain.
- Impulses are equally spaced every $1/\Delta T$.

Sampling

- FT of sampled functions

$$\tilde{f}(t) = f(t)S_{\Delta T}(t)$$

$$\tilde{F}(u) = \Im\{\tilde{f}(t)\} = \Im\{f(t)S_{\Delta T}(t)\} = F(u) \star S(u)$$

$$S(u) = \frac{1}{\Delta T}\sum_{N=-\infty}^{\infty} \delta(u - \frac{n}{\Delta T})$$

$$\begin{aligned}\tilde{F}(u) &= F(u) \star S(u) = \int_{-\infty}^{\infty} F(\tau)S(u-\tau)d\tau \\ &= \frac{1}{\nabla T}\int_{-\infty}^{\infty} F(\tau)\sum_{n=-\infty}^{\infty}\delta\left(u-\tau-\frac{n}{\Delta T}\right)d\tau \\ &= \frac{1}{\Delta T}\sum_{n=-\infty}^{\infty} F(u-\frac{n}{\Delta T})\end{aligned}$$

Sampling

- Sampling
  - The spectrum of the discrete signal consists of repetitions of the spectrum of the continuous signal every $1/\Delta T$.
  - The Nyquist criterion should be satisfied.

$$f(t) \leftrightarrow F(\mu)$$

$$\tilde{f}(n\Delta T) = f[n] \leftrightarrow \tilde{F}(\mu) = \frac{1}{\Delta T} \sum_{n=-\infty}^{+\infty} F\left(\mu - \frac{n}{\Delta T}\right)$$

Sampling

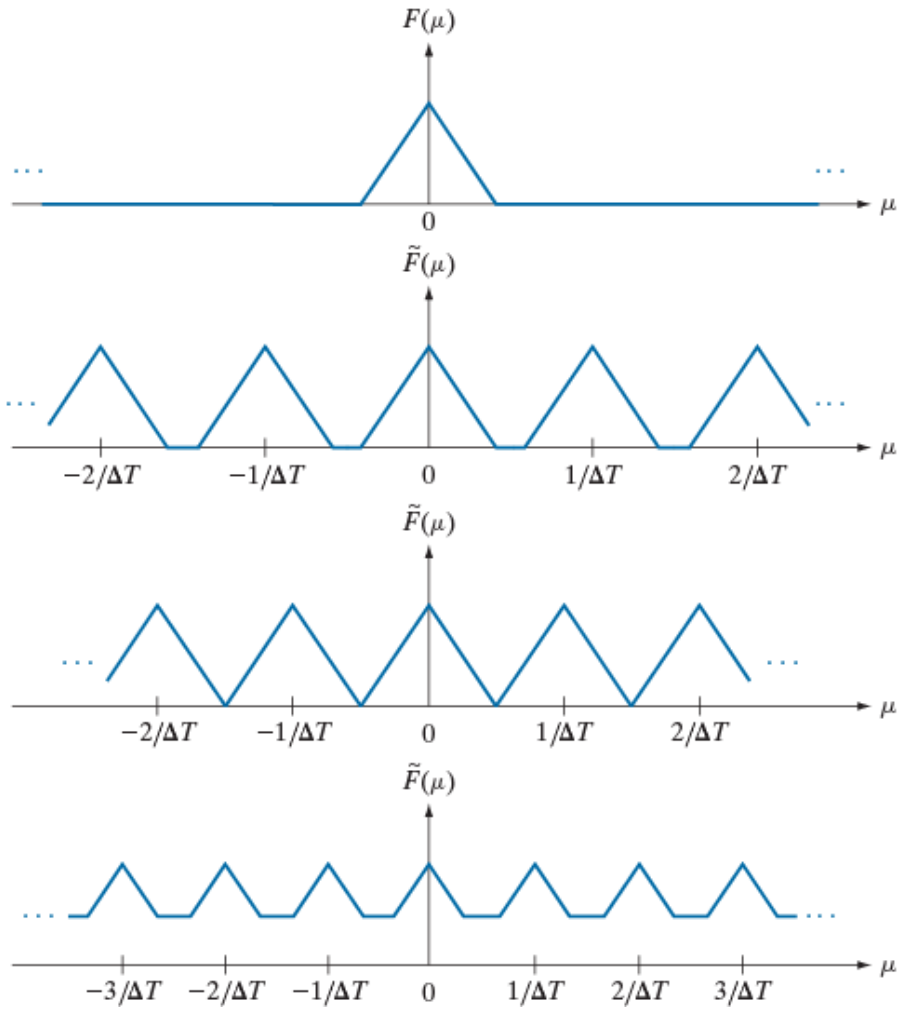
FT of a continuous signal

Oversampling

Critical sampling with the Nyquist frequency

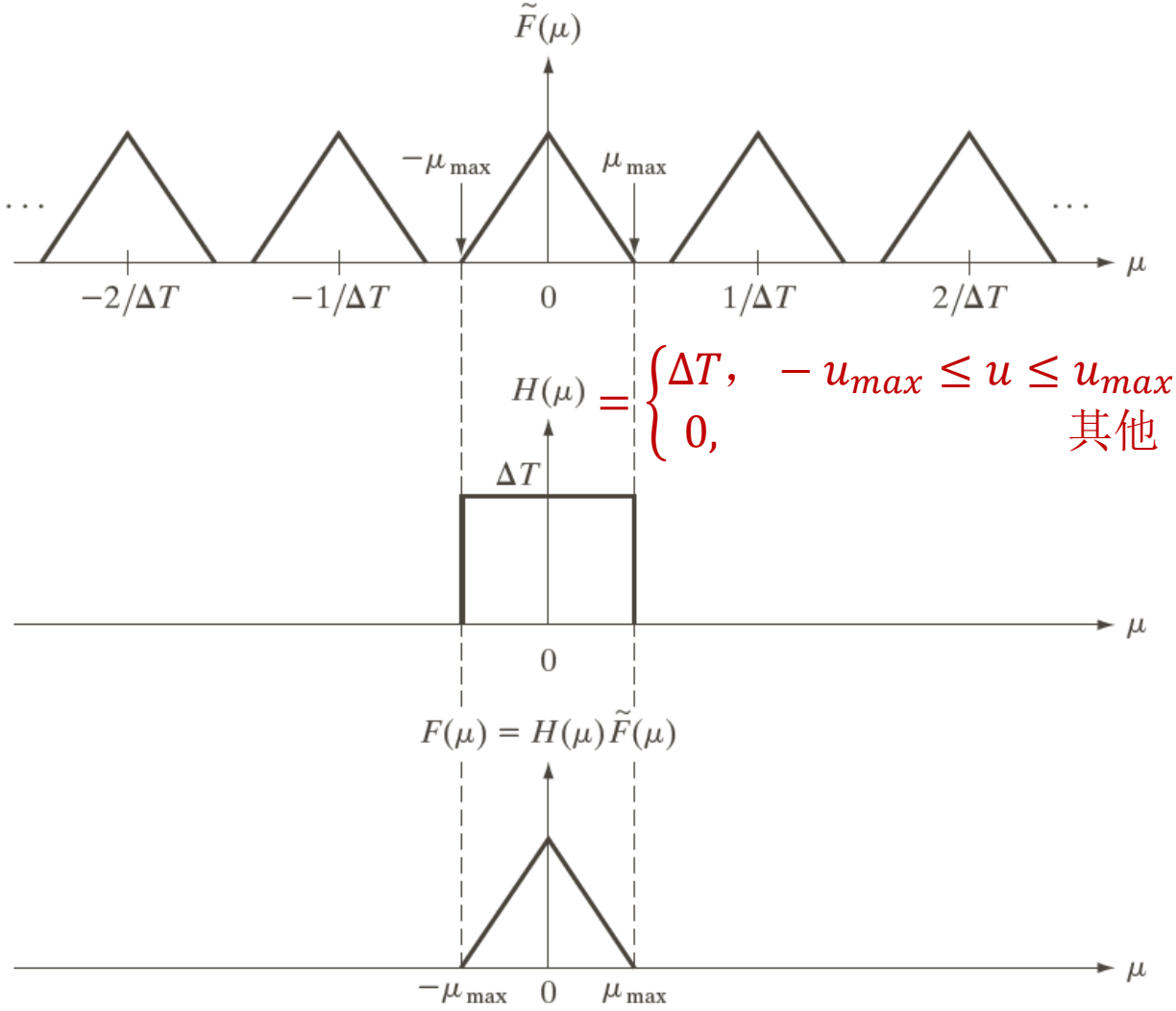
$$\frac{1}{\Delta T} \geq 2\mu_{\max}$$

Undersampling
Aliasing appears



Sampling

Recovery

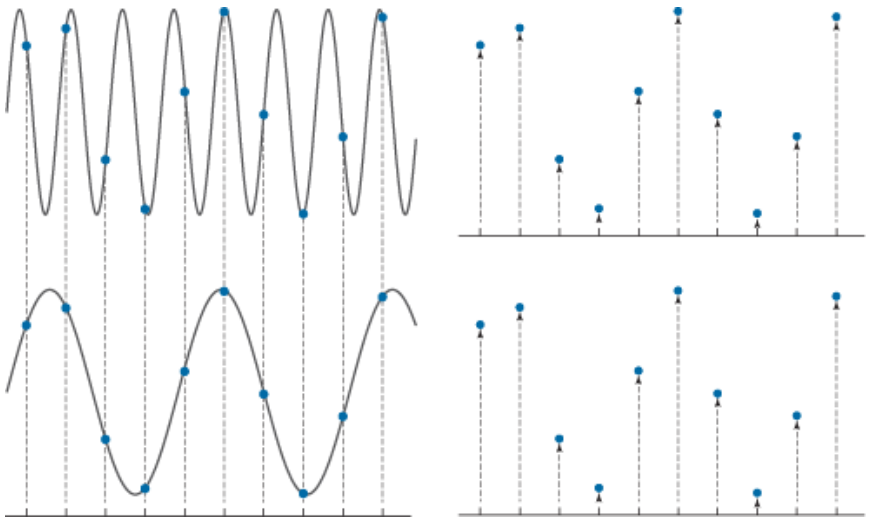


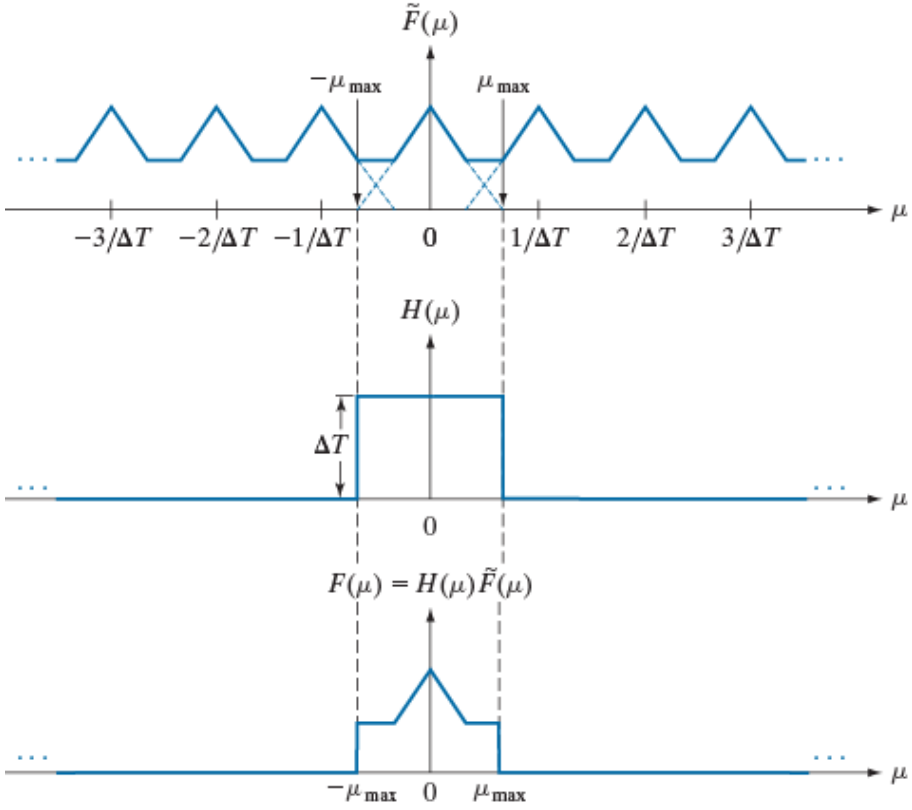
Ideal lowpass filter H(u):

Infinitely rapid transitions in amplitude, a characteristic that can not be achieved with physical electronic components.

Sampling

- Aliasing, the reconstruction of the continuous signal is not correct.



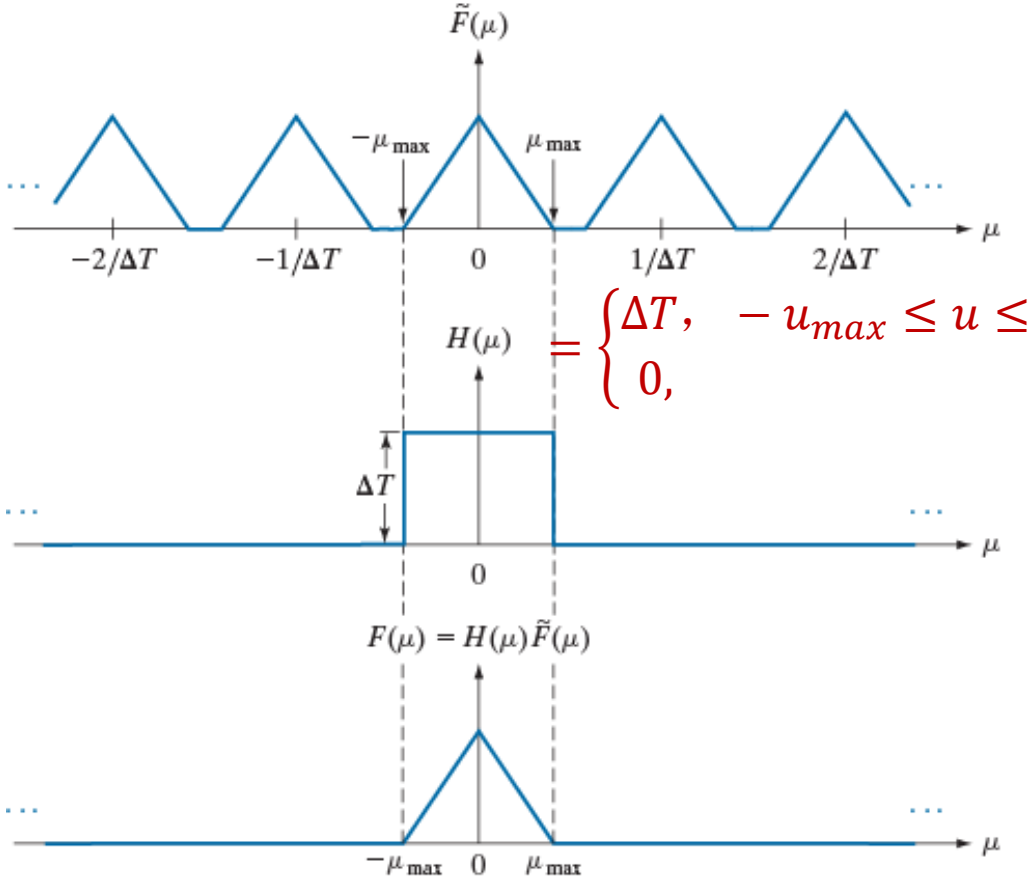


Sampling

- Reconstruction (correct sampling).

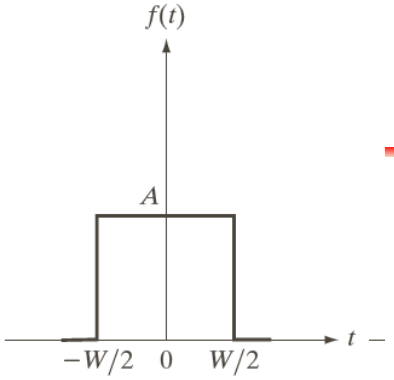
$$F(u) = H(u)\tilde{F}(u)$$

$$\begin{aligned} f(t) &= \mathfrak{F}^{-1}\{F(\mu)\} \\ &= \mathfrak{F}^{-1}\{H(\mu)\tilde{F}(\mu)\} \\ &= h(t) \star \tilde{f}(t) \end{aligned}$$



Sampling

- Reconstruction (correct s



$$f(t) = \Im^{-1}\{F(u)\} = \Im^{-1}\{H = AW\frac{sinc(\pi\mu W)}{(\pi\mu W)}$$

$$= h(t) \star \tilde{f}(t)$$

$$\tilde{f}(t) = f(t)S_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\Delta T)$$

$$= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} f(\tau)\delta(\tau - n\Delta T)h(t - \tau)d\tau$$

$$= \sum_{n=-\infty}^{\infty} f(n\Delta T)sinc[\frac{t - n\Delta T}{\Delta T}]$$

The perfect reconstruction function is the infinite sum of the sinc function weighted by the sampled value

Sampling

Reconstruction

- Provided a correct sampling, the continuous signal may be perfectly reconstructed by its samples.
- Perfectly reconstructed function is an infinite sum of sinc functions weighted by the sample values

$$f(t) = \sum_{n=-\infty}^{+\infty} f(n\Delta T)\ \mathrm{sinc}\left[\frac{(t - n\Delta T)}{n\Delta T}\right]$$

The Discrete Fourier Transform

- The Fourier transform of a sampled (discrete) signal is a continuous, periodic function of the frequency

$$\begin{aligned}\widetilde{F}(u) &= F(u) \star S(u) = \int_{-\infty}^{\infty} F(\tau) S(u-\tau) d\tau \\ &= \frac{1}{\nabla T} \int_{-\infty}^{\infty} F(\tau) \sum_{n=-\infty}^{\infty} \delta\left(u - \tau - \frac{n}{\Delta T}\right) d\tau \\ &= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} F\left(u - \frac{n}{\Delta T}\right)\end{aligned}$$

$$S(u) = \frac{1}{\Delta T} \sum_{N=-\infty}^{\infty} \delta\left(u - \frac{n}{\Delta T}\right)$$

- For a N-length discrete signal, taking N samples of its Fourier transform at frequencies:

$$\mu_k = \frac{k}{N\Delta T}, \; k = 0, 1, .., N-1$$

provides the discrete Fourier transform (DFT) of the signal.

The Discrete Fourier Transform

$$\tilde{F}(u) = \int_{-\infty}^{\infty} \tilde{f}(t) e^{-2j\pi\mu t} dt$$

$$= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\,\triangle\, T) e^{-2j\pi\mu t} dt$$

$$= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} f(t)\delta(t - n\,\triangle\, T) e^{-2j\pi\mu t} dt$$

$$= \sum_{n=-\infty}^{\infty} f_n e^{-j2\pi\mu n\triangle T}$$

$$\tilde{f}(t) = f(t) S_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\Delta T)$$

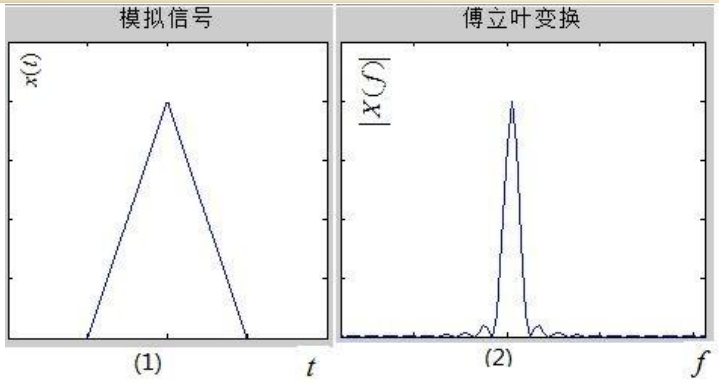
$$f_n = \int_{-\infty}^{\infty} f(t)\delta(t - n\,\triangle\, T) = f(n\,\triangle\, T)$$

$$\mu_k = \frac{k}{N\Delta T},\ k = 0, 1, .., N-1$$

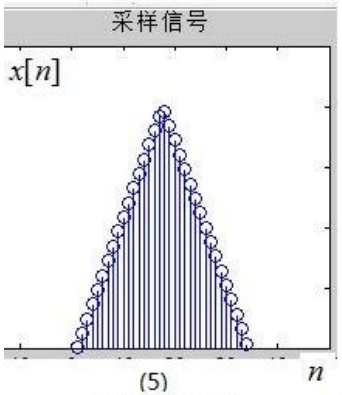
- DFT pair of signal f [n] of length N.

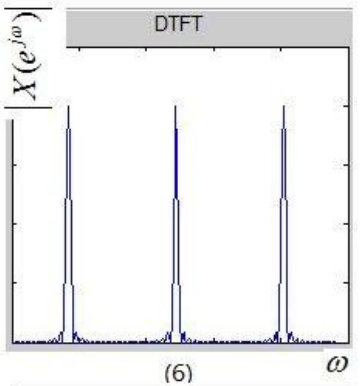
$$F[k] = \sum_{n=0}^{N-1} f[n] e^{-j\frac{2\pi nk}{N}}, \qquad 0 \le k \le N-1$$

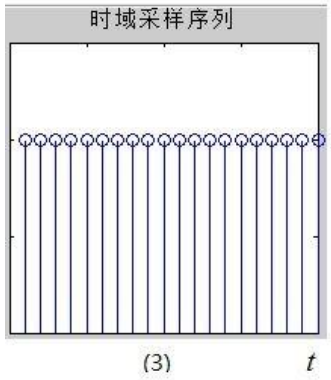
$$f[n] = \frac{1}{N} \sum_{n=0}^{N-1} F[k] e^{j\frac{2\pi nk}{N}}, \qquad 0 \le n \le N-1$$

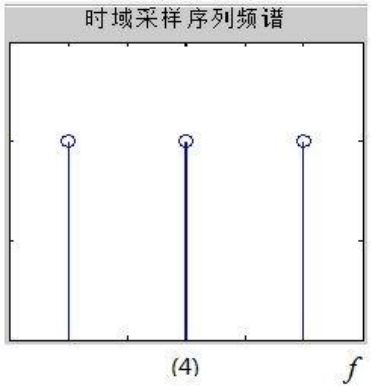

模拟信号
x(t)
(1)
t

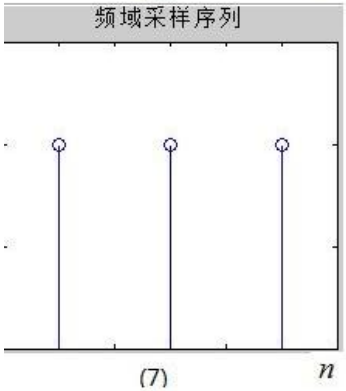
傅立叶变换
|X(f)|
(2)
f

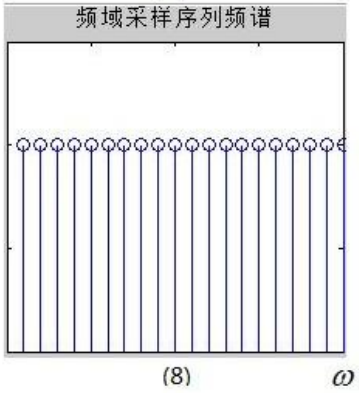

采样信号
x[n]
(5)
n

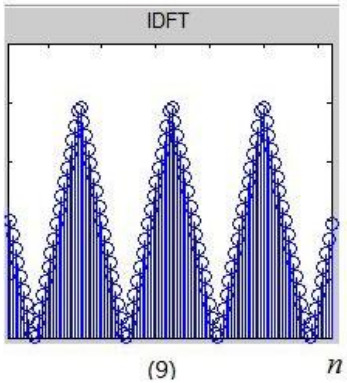

DTFT
|X(e^jω)|
(6)
ω

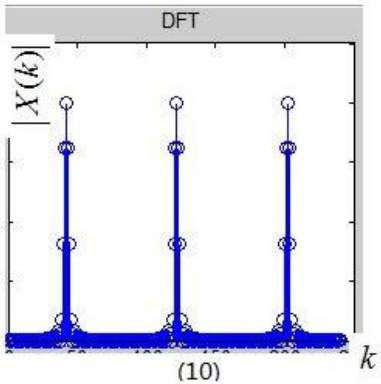

时域采样序列
(3)
t


时域采样序列频谱
(4)
f


频域采样序列
(7)
n


频域采样序列频谱
(8)
ω


IDFT
(9)
n


DFT
|X(k)|
(10)
k

The Discrete Fourier Transform

Property

- The DFT of a N-length $f[n]$ signal is periodic with period N

$$F[k+N] = F[k]$$

- This is due to the periodicity of the complex exponential:

$$\left(w_N\right)^n = \left(e^{-j\frac{2\pi}{N}}\right)^n \Leftrightarrow w_N^n = e^{-j\frac{2\pi n}{N}}$$

Relationship Between Sampling and Frequency Intervals

- If $f(x)$ consists of M samples of a function $f(t)$ taken ΔT units apart, the duration of the record comprising the set $\{f(x)\}, x = 0,1,2,\ldots,M-1$, is

$$T = M\Delta T$$

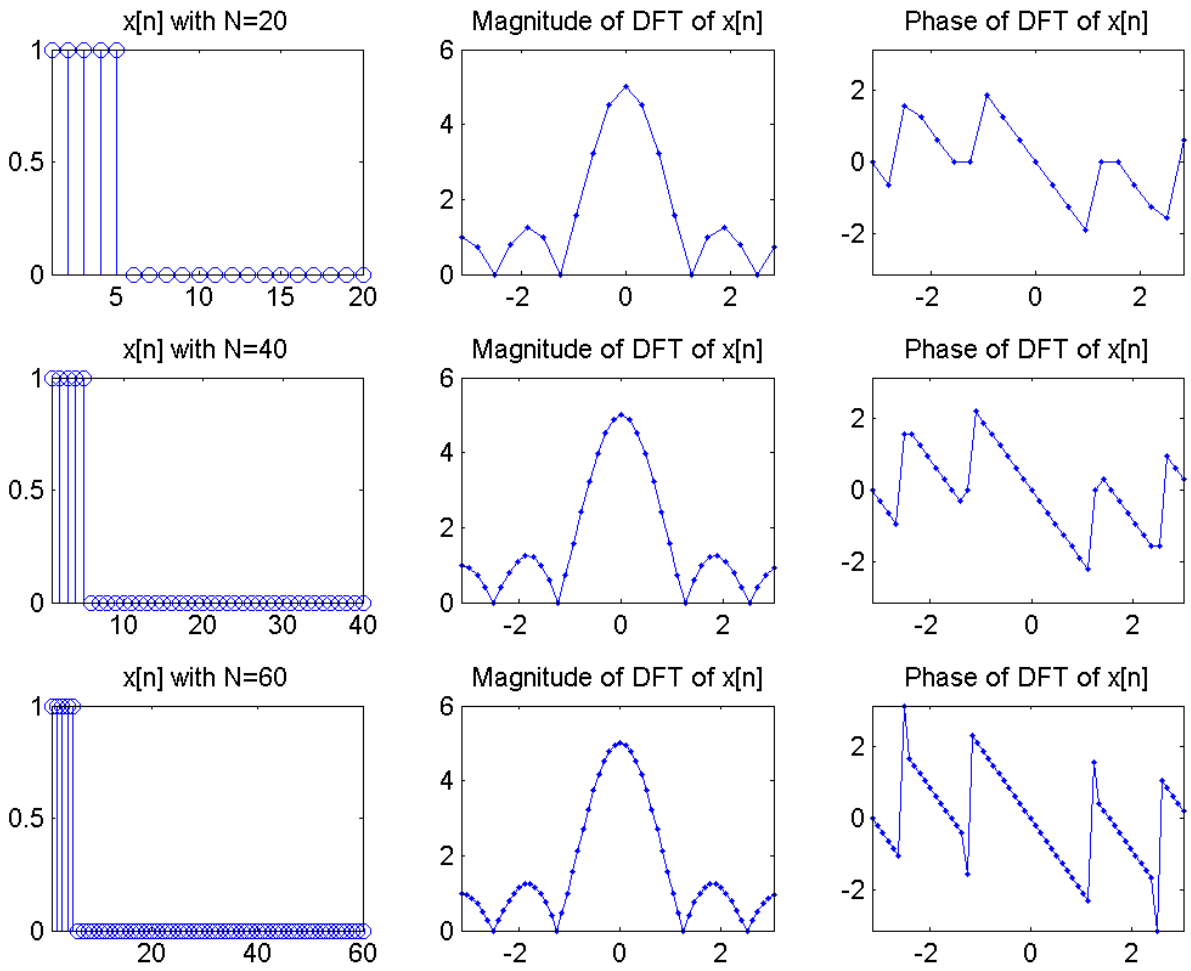
- The corresponding spacing, $\Delta\mu$, in the discrete frequency domain follows:

$$\Delta\mu = \frac{1}{M\Delta T} = \frac{1}{T}$$

- The entire frequency range spanned by the M components of the DFT is:

$$\Omega = M\Delta\mu = \frac{1}{\Delta T}$$

Zero Padding



Worked Example: Zero Padding

Sampling rate: 100 Hz; number of samples: 100.

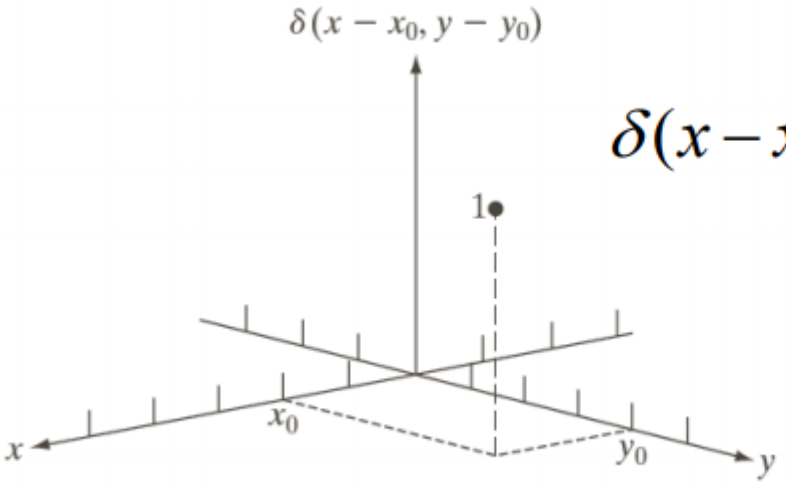
Original DFT grid: $\Delta f = 100/100 = 1$ Hz.

Pad to 1000 points: $\Delta f_grid = 0.1$ Hz.

The spectrum is sampled more densely; no new data are added.

Question: did the observation duration increase?

2D Continuous signals

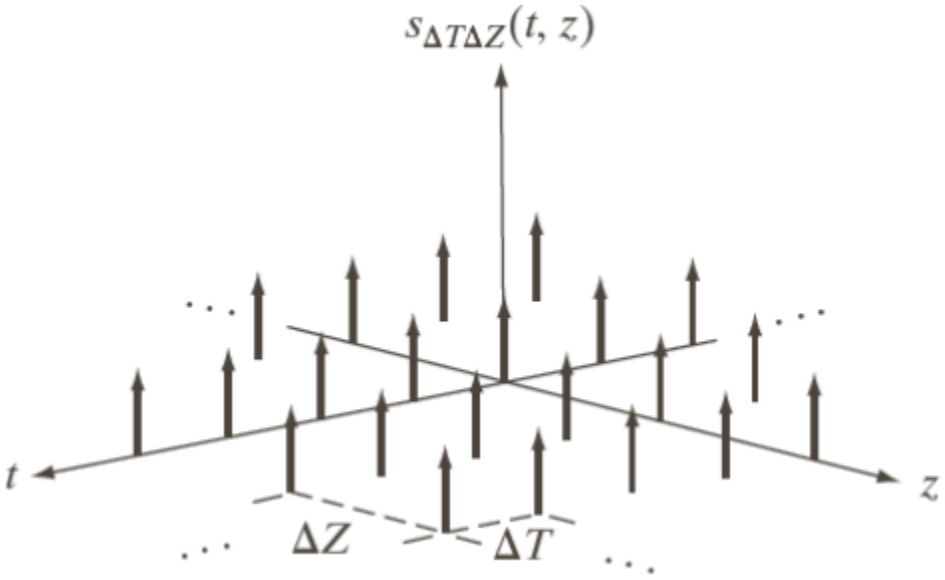


$$\delta(x-x_0,y-y_0)=\begin{cases}+\infty, & x=x_0,y=y_0\\ 0 & \text{otherwise}\end{cases}$$

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)\delta(x-x_0,y-y_0)dydx = f(x_0,y_0)$$

Separable: $\delta(x-x_0,y-y_0)=\delta(x-x_0)\delta(y-y_0)$

2D Continuous signals



The 2D impulse train is also separable:

$$S_{\Delta X \Delta Y}(x, y) = S_{\Delta X}(x) S_{\Delta Y}(y) = \sum_{n=-\infty}^{+\infty} \sum_{m=-\infty}^{+\infty} \delta(x - n\Delta X, y - n\Delta Y)$$

2D Continuous signals

- The Fourier transform of a continuous 2D signal *f(x,y)*.

$$F(\mu,\nu)=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)e^{-j2\pi(\mu x+\nu y)}dydx$$

$$f(x,y)=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} F(\mu,\nu)e^{j2\pi(\mu x+\nu y)}d\nu d\mu$$

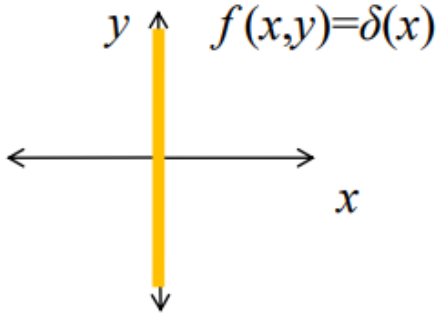
2D Continuous signals

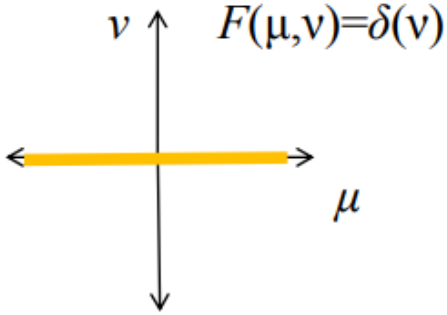
- Example: FT of $f(x,y)=\delta(x)$

$$F(\mu,\nu)=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\delta(x)e^{-j2\pi(\mu x+\nu y)}dydx$$

$$=\int_{-\infty}^{+\infty}\delta(x)e^{-j2\pi\mu x}dx\int_{-\infty}^{+\infty}e^{-j2\pi\nu y}dy$$

$$=\int_{-\infty}^{+\infty}e^{-j2\pi\nu y}dy\ =\delta(\nu)$$





2D Continuous signals

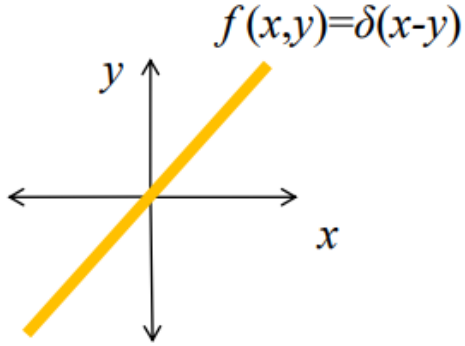
- Example: FT of $f(x,y)=\delta(x-y)$

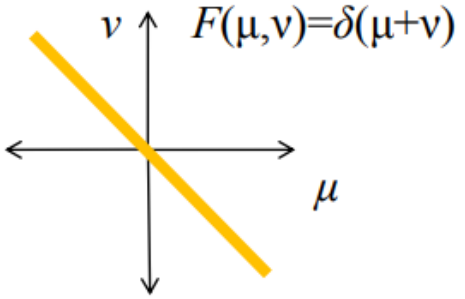
$$F(\mu,\nu)=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\delta(x-y)e^{-j2\pi(\mu x+\nu y)}dydx$$

$$=\int_{-\infty}^{+\infty}\left[\int_{-\infty}^{+\infty}\delta(x-y)e^{-j2\pi\mu x}dx\right]e^{-j2\pi\nu y}dy$$

$$=\int_{-\infty}^{+\infty}e^{-j2\pi\mu y}e^{-j2\pi\nu y}dy=\int_{-\infty}^{+\infty}e^{-j2\pi(\mu+\nu)y}dy$$

$$=\delta(\mu+\nu)$$







Check: Stripe Direction

Vertical stripes vary along the horizontal axis.

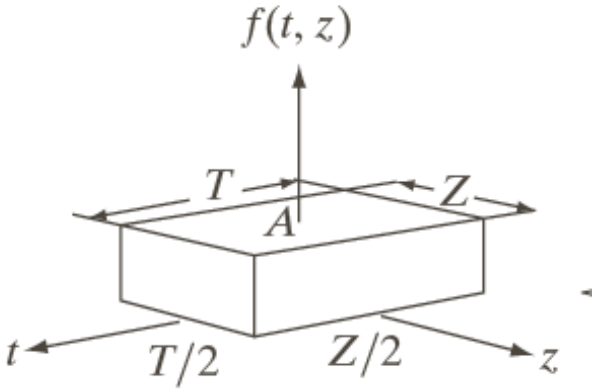
Their sinusoidal peaks lie on the horizontal frequency axis.

Closer stripes → higher frequency → peaks farther from DC.

Question: what happens when the stripes rotate by 45° ?

2D Continuous signals

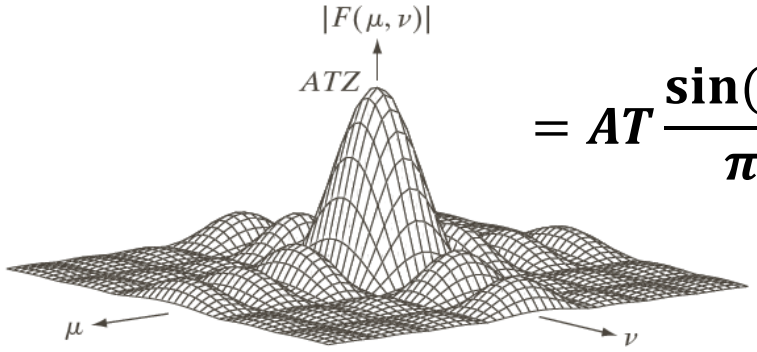
$$F(\mu,v)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}f(t,z)e^{-j2\pi(\mu t+vz)}dtdz$$



$$=\int_{-\frac{T}{2}}^{\frac{T}{2}}\int_{-\frac{Z}{2}}^{\frac{Z}{2}}Ae^{-j2\pi(\mu t+vz)}dtdz$$

$$=A\int_{-\frac{T}{2}}^{\frac{T}{2}}e^{-j2\pi\mu t}dt\int_{-\frac{Z}{2}}^{\frac{Z}{2}}e^{-j2\pi vz}dz$$

$$=\frac{-A}{j2\pi\mu}\left[e^{-j2\pi\mu T}-e^{j2\pi\mu T}\right]\frac{-1}{j2\pi v}\left[e^{-j2\pi vZ}-e^{j2\pi vZ}\right]$$



$$=AT\frac{\sin(\pi\mu T)}{\pi\mu T}Z\frac{\sin(\pi vZ)}{\pi vZ}=ATZ\left[\frac{\sin(\pi\mu T)}{\pi\mu T}\right]\left[\frac{\sin(\pi vZ)}{\pi vZ}\right]$$

2D continuous signals

- 2D continuous convolution

$$f(x,y)*h(x,y)=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x-\alpha,y-\beta)h(\alpha,\beta)d\alpha d\beta$$

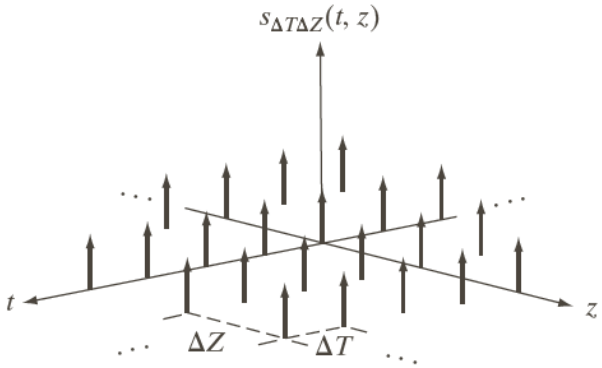
- Convolution property

$$f(x,y)*h(x,y) \leftrightarrow F(\mu,\nu)H(\mu,\nu)$$

2D continuous signals

- 2D sampling is accomplished by

$$S_{\Delta X \Delta Y}(x, y) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(x - n\Delta X, y - n\Delta Y)$$



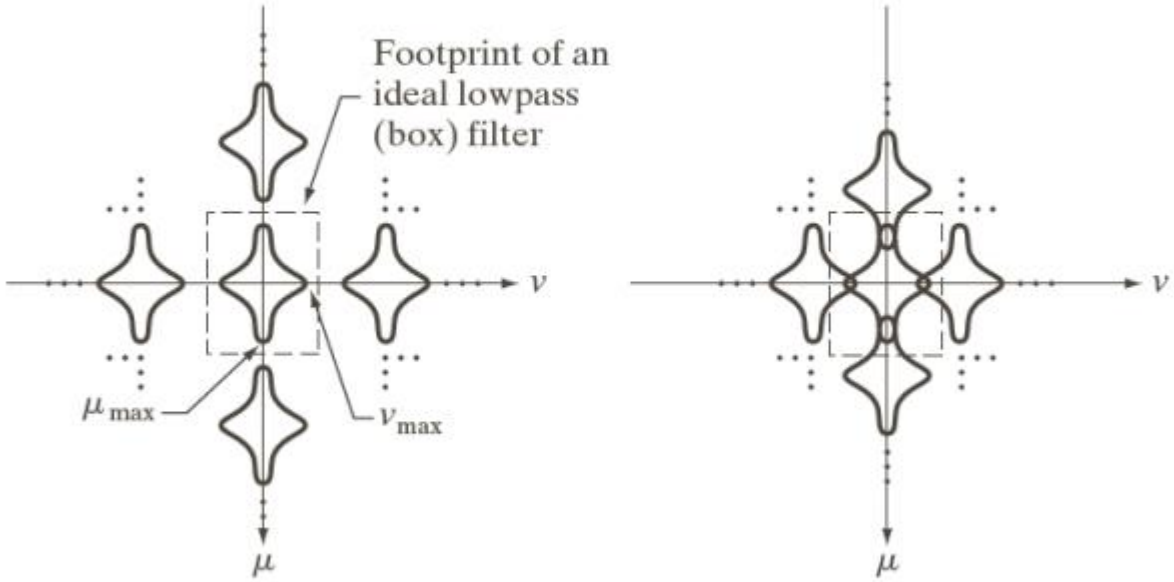
- The FT of the sampled 2D signal consists of repetitions of the spectrum of the 1D continuous signal.

$$\tilde{F}(\mu,\nu) = \frac{1}{\Delta X}\frac{1}{\Delta Y}\sum_{m,n=-\infty}^{+\infty} F\left(\mu - \frac{m}{\Delta X}, \nu - \frac{n}{\Delta Y}\right)$$

2D continuous signals

- The Nyquist theorem involves both the horizontal and vertical frequencies.

$$\frac{1}{\Delta X} \geq 2\mu_{\max}, \quad \frac{1}{\Delta Y} \geq 2v_{\max}$$



Over-sampled

Under-sampled

Aliasing

A continuous function of two continuous variables, t and z, can be band-limited in general only if it extends infinitely in both coordinate directions. The very act of limiting the duration of the function introduces corrupting frequency components extending to infinity in the frequency domain.

- ✓ Because we cannot sample a function infinitely, aliasing is always present in digital images.
- ✓ Two principal manifestations of aliasing in images: ***spatial aliasing*** and ***temporal aliasing***
  - ✓ **Spatial aliasing**：Due to under-sampling
  - ✓ **Temporal Aliasing**：Related to time intervals between images in a sequences of images. “wagon wheel”
- ✓ Anti-aliasing filtering：Slightly defocusing the scene to be digitized
  - ✓ Anti-aliasing filtering has to be done at the “front-end,”, before the image is samples

Aliasing

二维情况下，若只在两个坐标方向上无限扩展，二维连续变量t和z的连续函数通常可以被带限。而限制一个函数时间的非常动作则将引起在频率域无限扩展的有害的频率分量。

- ✓ 图像取样中不可能做无限取样，因此取样图像中无可避免地存在混淆。
- ✓ 图像中，存在两种主要的混淆现象：**空间混淆**和**时间混淆**。
  - ✓ 空间混淆：欠采样导致
  - ✓ 时间混淆：图像序列中图像间的时间间隔有关。“车轮效应”
- ✓ 抗混淆滤波：在图像被取样之前通过稍微散焦来削弱高频
  - ✓ 发生在图像被取样之前，反混淆滤波常嵌入在镜头或传感器上，因此很难通过图像去说明

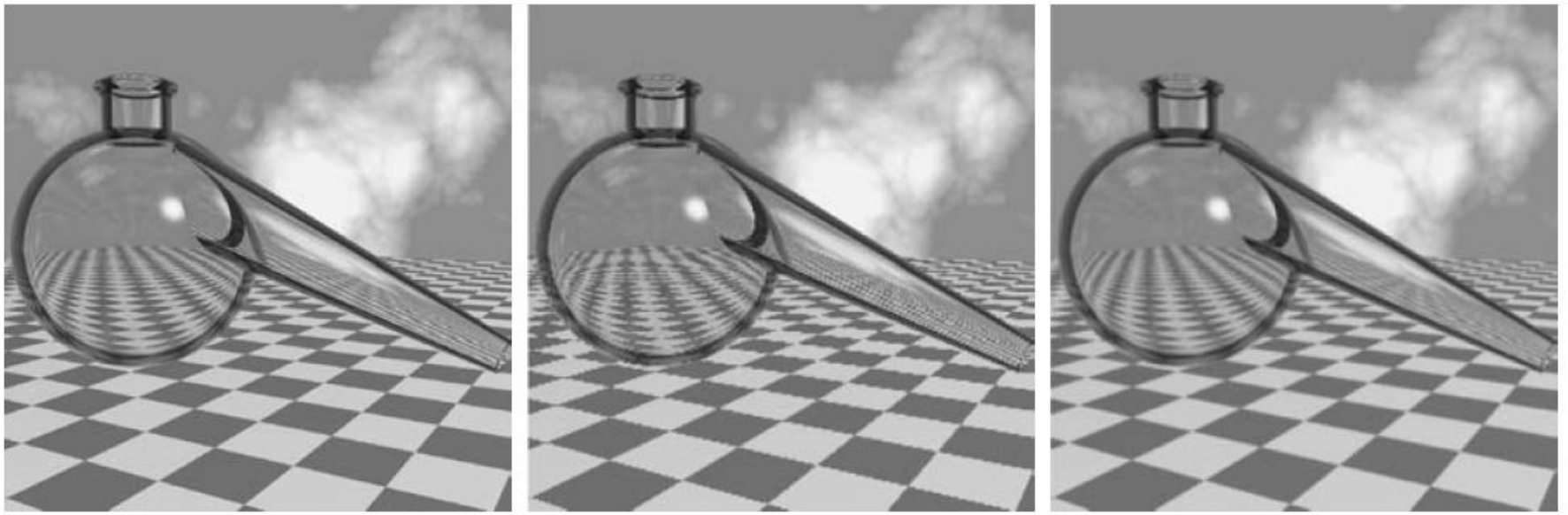
Image Interpolation and Resampling



a b c

FIGURE 4.19 Illustration of aliasing on resampled natural images. (a) A digital image of size 772×548 pixels with visually negligible aliasing. (b) Result of resizing the image to 33% of its original size by pixel deletion and then restoring it to its original size by pixel replication. Aliasing is clearly visible. (c) Result of blurring the image in (a) with an averaging filter prior to resizing. The image is slightly more blurred than (b), but aliasing is not longer objectionable. (Original image courtesy of the Signal Compression Laboratory, University of California, Santa Barbara.)

Image Interpolation and Resampling



a b c

FIGURE 4.18 Illustration of jaggies. (a) A 1024 × 1024 digital image of a computer-generated scene with negligible visible aliasing. (b) Result of reducing (a) to 25% of its original size using bilinear interpolation. (c) Result of blurring the image in (a) with a 5 × 5 averaging filter prior to resizing it to 25% using bilinear interpolation. (Original image courtesy of D. P. Mitchell, Mental Landscape, LLC.)



Case Study: Shrinking a Checkerboard

Reduce image width and height by a factor of 4.

Direct decimation may create false large-scale patterns.

Low-pass filtering before decimation suppresses aliasing.

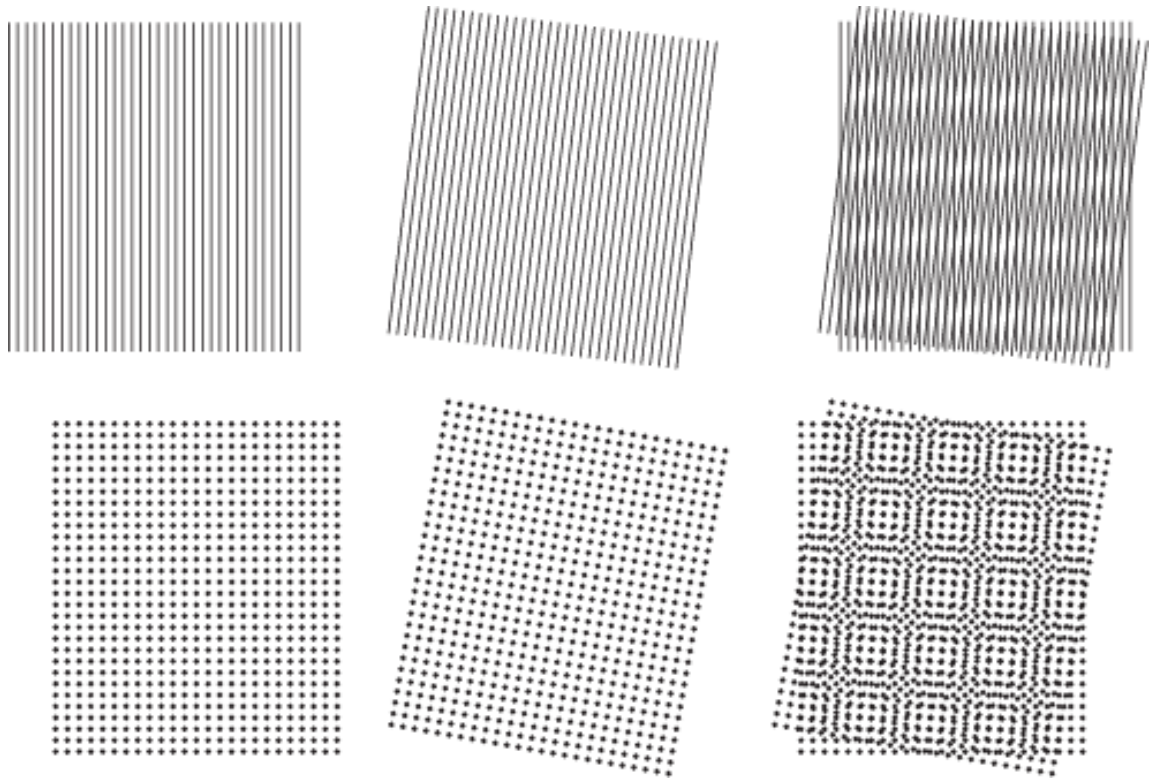
Question: why can a slightly blurred input give a better result?

Aliasing - Moiré Patterns

- Effect of sampling a scene with periodic or nearly periodic components (e.g. overlapping grids, TV raster lines and stripped materials).
- In image processing the problem arises when scanning media prints (e.g. magazines, newspapers).
- The problem is more general than sampling artifacts.

Aliasing - Moiré Patterns

- Superimposed grid drawings (not digitized) produce the effect of new frequencies not existing in the original components.



a b c
d e f

FIGURE 4.20
Examples of the moiré effect. These are vector drawings, not digitized patterns. Superimposing one pattern on the other is analogous to multiplying the patterns.

Aliasing - Moiré Patterns

- In the printing industry the problem comes when scanning photographs from the superposition of:
  - The sampling lattice (usually horizontal and vertical).
  - Dot patterns on the newspaper image.

The 2D DFT

- 2D DFT pair of image f [m,n] of size MxN

$$F[k,l] = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f[m,n] e^{-j2\pi\left(\frac{km}{M}+\frac{ln}{N}\right)}$$

$$f[m,n] = \frac{1}{MN}\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} F[k,l] e^{j2\pi\left(\frac{km}{M}+\frac{ln}{N}\right)}$$

$$\begin{cases} 0 \le k \le M-1 \\ 0 \le l \le N-1 \end{cases}, \qquad \begin{cases} 0 \le m \le M-1 \\ 0 \le n \le N-1 \end{cases}$$

The 2D DFT

Relationships Between Spatial and Frequency Intervals

- Continuous function *f(t,z)* is sampled to form a digital image, *f(x,y)*, consisting of $M \times N$ samples taken in the t- and z-directions, respectively.

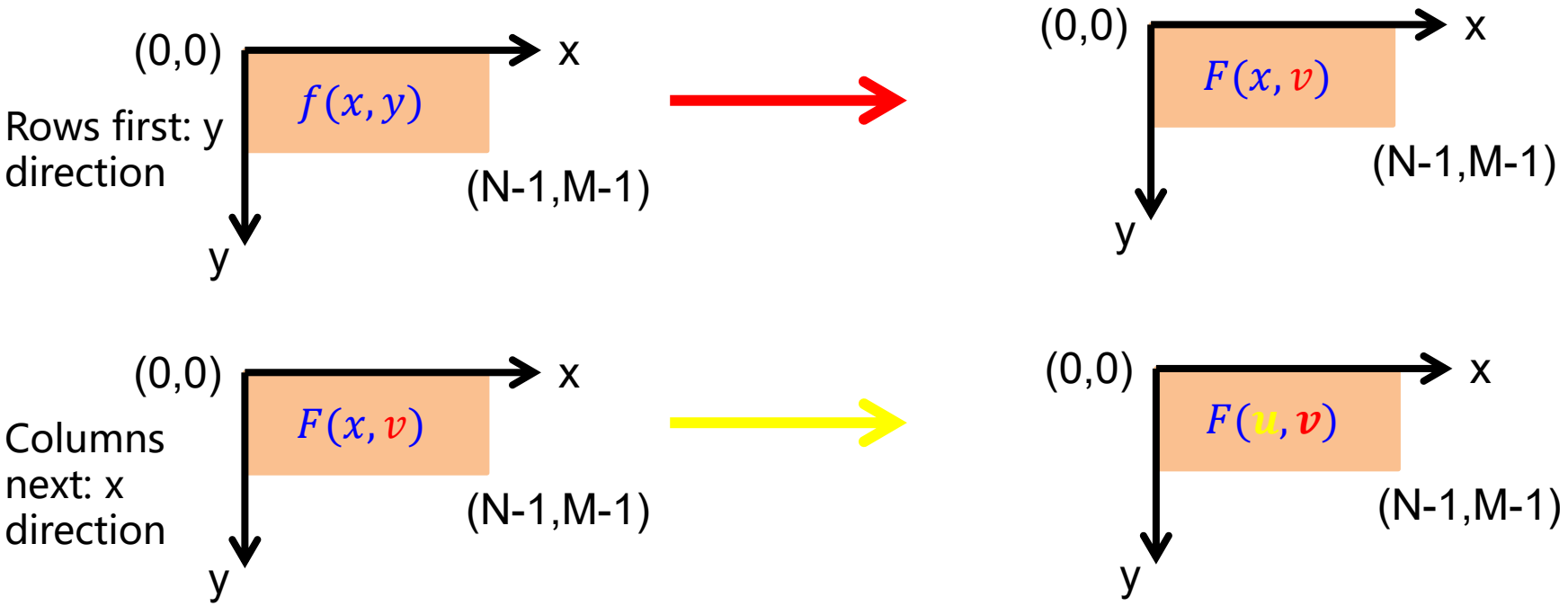
$$\Delta u = \frac{1}{M\Delta T}, \Delta v = \frac{1}{N\Delta Z}$$

Note that the separations between samples in the frequency domain are inversely proportional both to the spacing between spatial samples and the number of samples.

The 2D DFT

Separability

- The 2D DFT can be computed by computing 1D DFT transforms along the rows (columns) of the image, followed by 1D transforms along the columns (rows) of the result.



The 2D DFT

Translation

- Multiplying $f(x, y)$ by the exponential shifts the origin of the DFT to (u_0, v_0)

$$f(x,y)e^{j2\pi(\frac{u_0 x}{M}+\frac{v_0 y}{N})} \Leftrightarrow F(u-u_0, v-v_0)$$

- Multiplying $F(u, v)$ by the negative of that exponential shifts the origin of $f(x, y)$ to (x_0, y_0).

$$F\{f(x-x_0, y-y_0)\} = e^{-j2\pi(\frac{x_0 u}{M}+\frac{y_0 v}{N})}F(u,v)$$

The 2D DFT

Rotation

- Rotating $f(x,y)$ by an angle θ_0 rotates $F(u,v)$ by the same angle. Conversely, rotating $F(u,v)$ rotates $f(x,y)$ by the same angle.

$$f(r,\theta+\theta_0) \Leftrightarrow F(\rho,\varphi+\theta_0)$$

Mean

- Zero-frequency term (dc component) is proportional to the average value of $f(x,y)$

$$F(0,0) = MN \cdot \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(\frac{0*u}{M}+\frac{0*v}{N}\right)} = MN \cdot \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)$$

$$= MN \cdot \bar{f}(x,y)$$

MN is usually large, typically is the largest component of the spectrum by a factor that can be several orders of magnitude larger than other terms.

The 2D DFT

Periodicity

- As in the 1D case, the 2D DFT and its inverse are infinitely periodic in the *u* and *v* directions; that is,

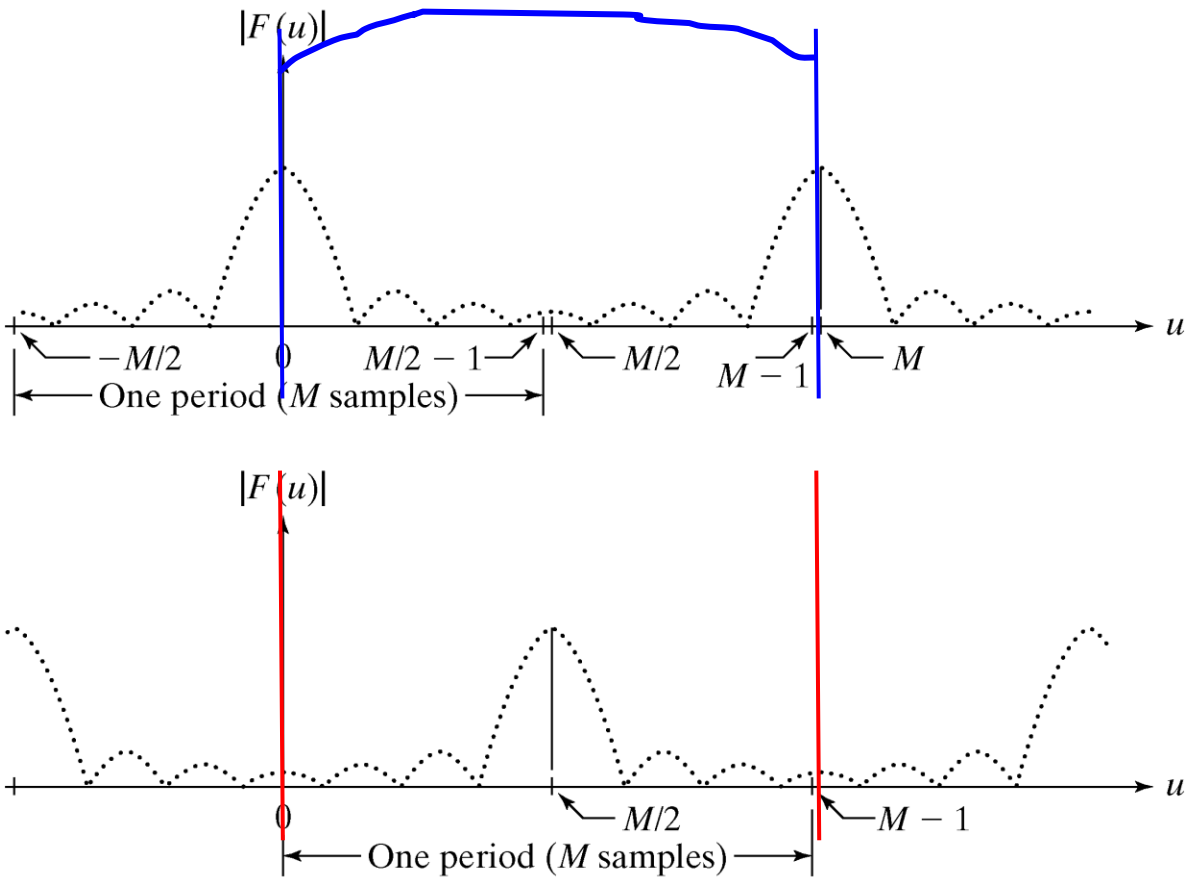
$$F(u,v) = F(u + k_1 M, v) = F(u, v + k_2 N) = F(u + k_1 M, v + k_2 N)$$

$$f(x,y) = f(x + k_1 M, y) = f(x, y + k_2 N) = f(x + k_1 M, y + k_2 N)$$

where k_1 and k_2 are integers.

.

The 2D DFT



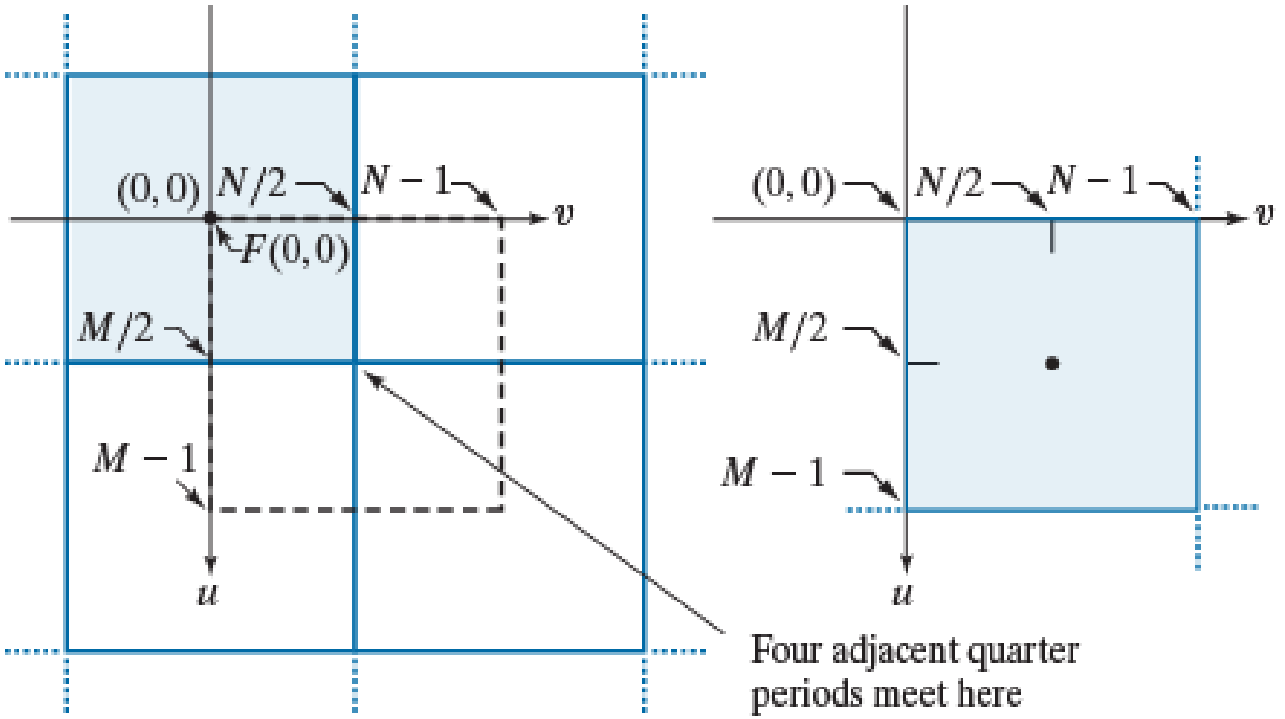
DFT output starts with the zero-frequency bin.

fftshift places zero frequency at the center.

For even N, multiplying by (−1)^x shifts the DFT by N/2 bins.

$$f(x)(-1)^x \Leftrightarrow F(u - M/2)$$

The 2D DFT



In two dimensions: $f[m,n](-1)^{m+n} \Leftrightarrow F(k-M/2, l-N/2)$

and $F(0,0)$ is now located at $(M/2, N/2)$.

The 2D DFT

- Fourier Spectrum and Phase Angle
  - In polar form: $\boldsymbol{F(u,v) = |F(u,v)|e^{j\varphi(u,v)}}$
  - Magnitude (Fourier or frequency spectrum):

$$|\boldsymbol{F(u,v)}| = \left[\boldsymbol{R^2(u,v) + I^2(u,v)}\right]^{\boldsymbol{1/2}}$$

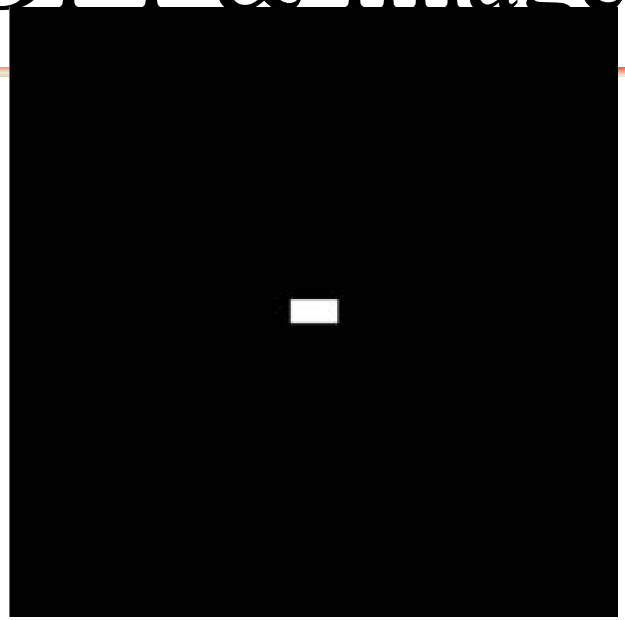
  - Phase angle:

$$\varphi(u,v) = arctan\left[\frac{I(u,v)}{R(u,v)}\right]$$

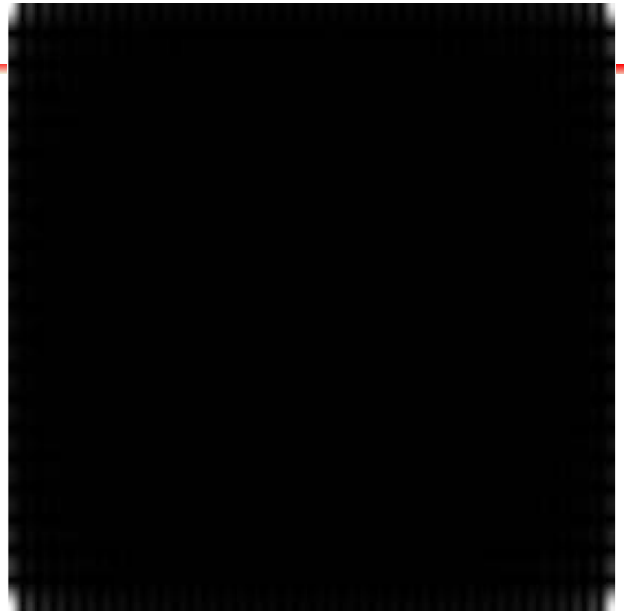
  - **Power spectrum:**

$$P(u,v) = |F(u,v)|^2 = [R^2(u,v) + I^2(u,v)]$$

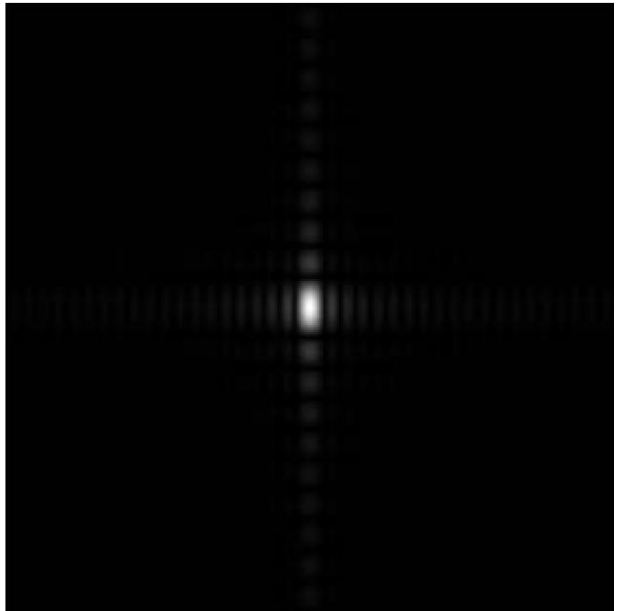
DFT & Images



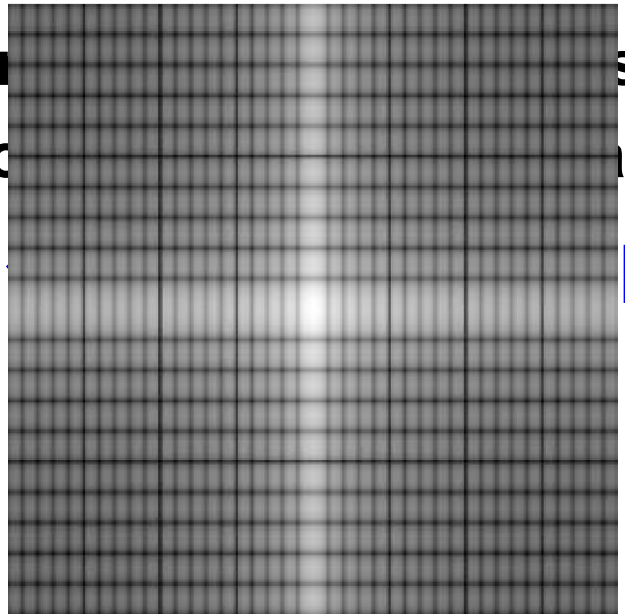
The origins of both the spatial and frequency domains are at the top left.



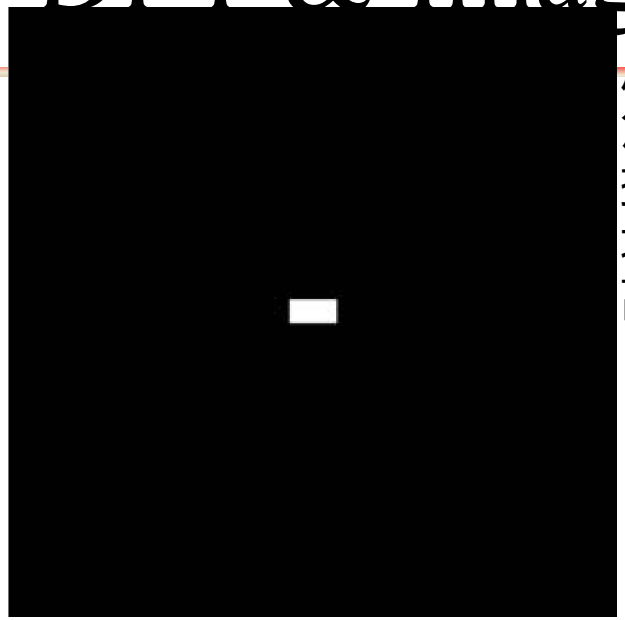
To center the spectrum, we simply myltiply the image by $(-1)^{x+y}$ before computing DFT



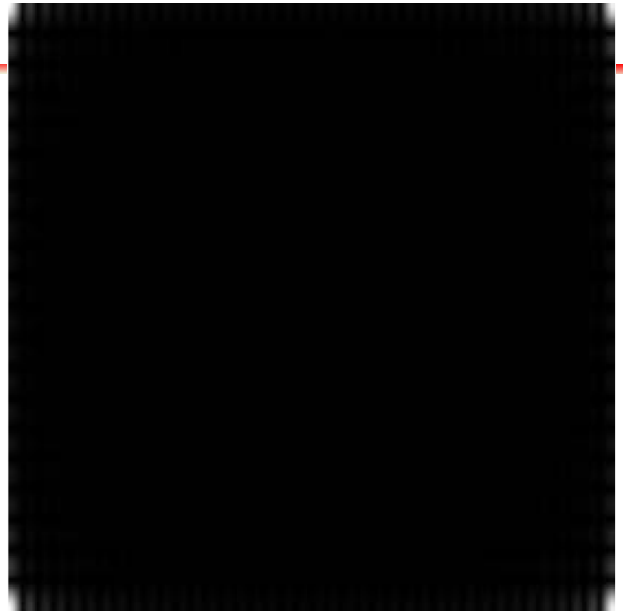
- **To b… s, we perfo… tion:**
- $D(u,$ …$|)$



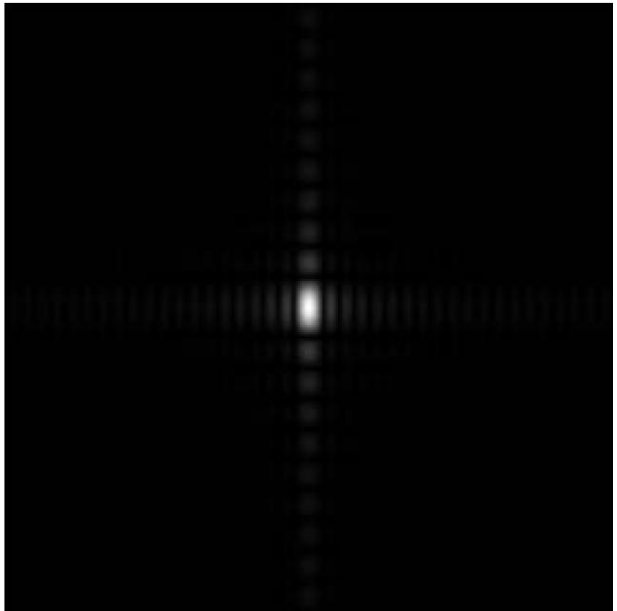
DFT & Images



原点在左上角，所以变换的原点区域包含了最高值，



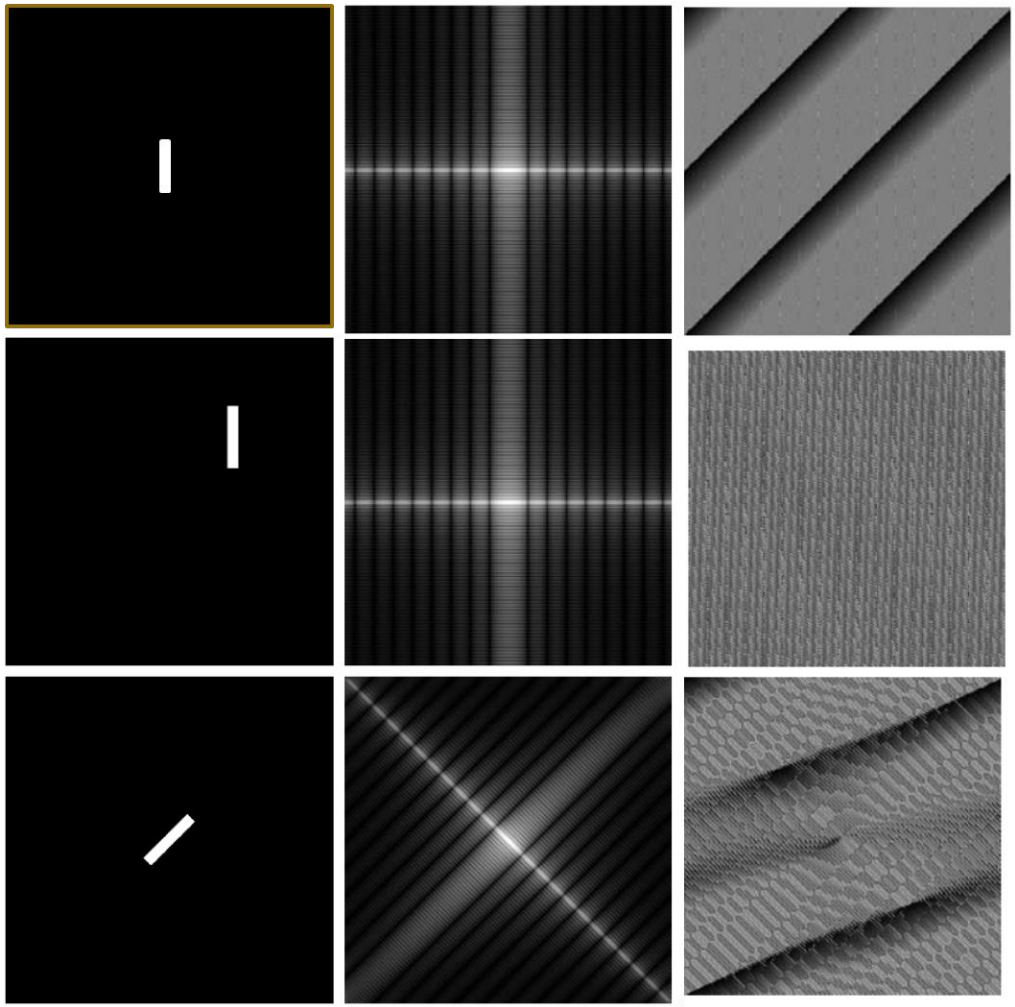
周期性中心化该谱，变换前原图像乘以 $(-1)^{x+y}$



- **因为人的视觉可分辨的灰度有限，频谱的显示都要取log变换**
- $D(u,v) = log(1 + |F(u,v)|)$

DFT & Images

- Fourier Spectrum is insensitive to image translation, but it rotates by the same angle of a rotated image.



$$F(u,v) = |F(u,v)|e^{j\varphi(u,v)}$$

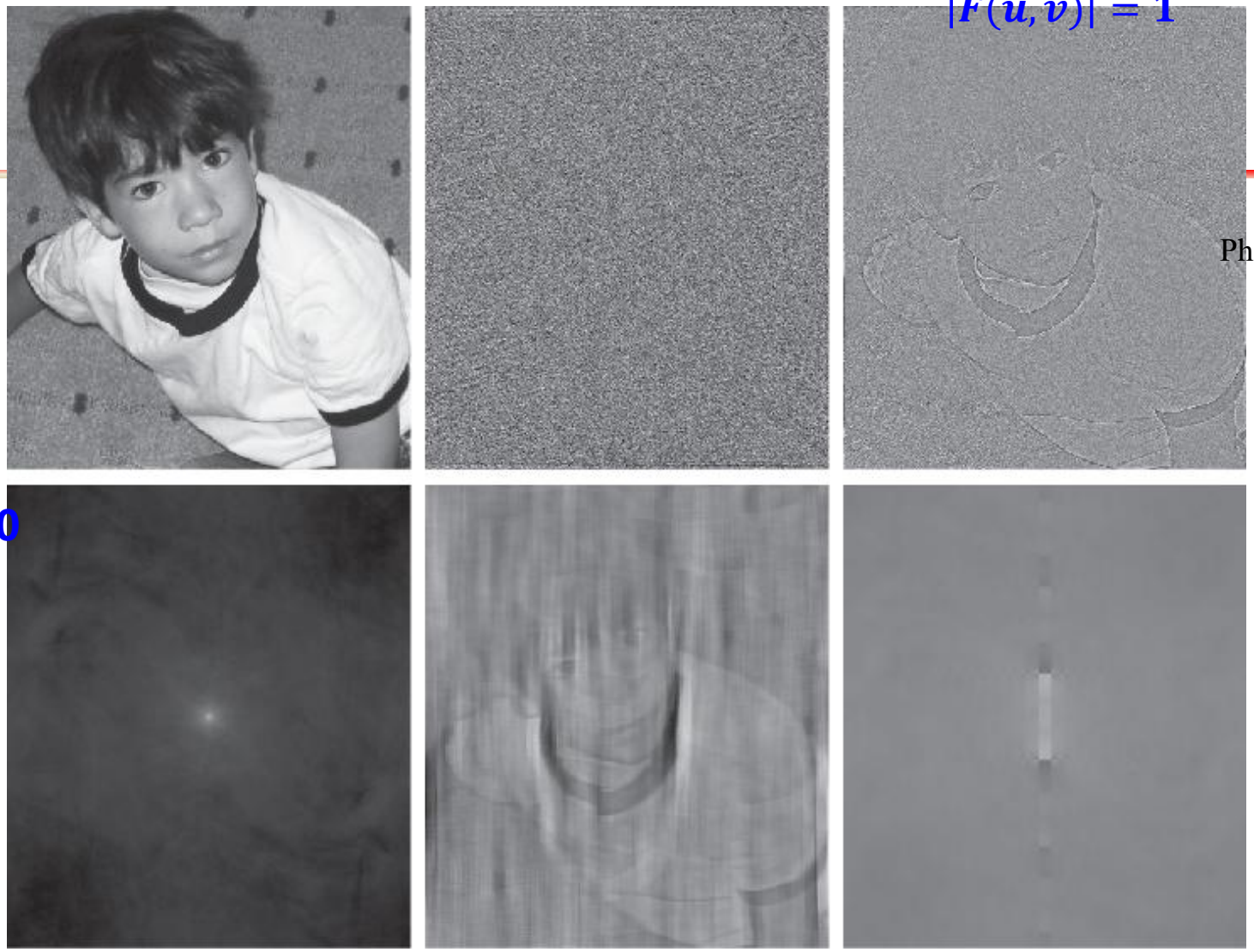
The 2D DFT

Fourier Spectrum and Phase Angle

- The component of spectrum of the DFT determine the amplitudes of the sinusoids that combine to form the resulting image.
- The phase is a measure of displacement of the various sinusoids with respect to their origin.
- The magnitude of the 2D DFT is an array whose components determine the intensities in the image. The corresponding phase is an array of angles that carry much of the information about where discernable objects are located in the image.

The 2D DFT

- Fourier Spectrum and Phase Angle
- DFT的谱的分量决定正弦波的幅度，这些正弦波结合起来可形成结果图像。在一幅图像的DFT中，在任何给定的频率处，较大的幅度意味着图像中该频率的正弦波比较突出。反之，较小的幅度则意味着图像中出现的正弦波比较少。
- 当二维DFT的幅度是一个阵列时，其分量决定了图像中的灰度，相应的相角则是一个角度的阵列，携带较多的关于图像中可辨别的物体定位的信息。



a b c
d e f

FIGURE 4.26 (a) Boy image. (b) Phase angle. (c) Boy image reconstructed using only its phase angle (all shape features are there, but the intensity information is missing because the spectrum was not used in the reconstruction). (d) Boy image reconstructed using only its spectrum. (e) Boy image reconstructed using its phase angle and the spectrum of the rectangle in Fig. 4.23(a). (f) Rectangle image reconstructed using its phase and the spectrum of the boy's image.

2D Convolution Theorem

- 2D circular convolution:

$$F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(\frac{ux}{M}+\frac{vy}{N})}$$

$$f(x,y) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(\frac{ux}{M}+\frac{vy}{N})}$$

2D Convolution Theorem

$$f(x,y) \star h(x,y) \Leftrightarrow F(u,v)H(u,v)$$

$$f(x,y)h(x,y) \Leftrightarrow \frac{1}{MN}F(u,v) \star H(u,v)$$

Convolution

DFT product

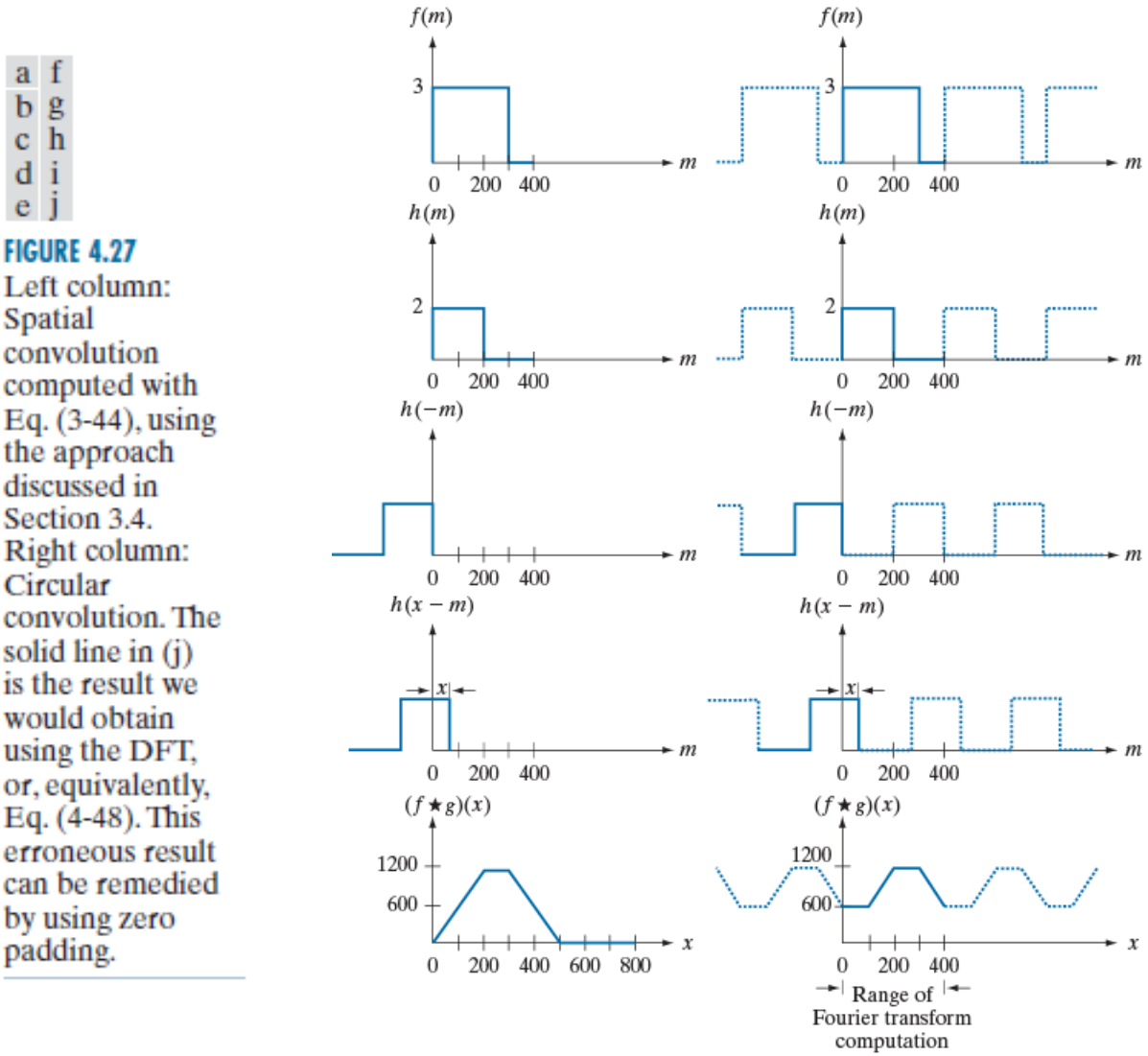


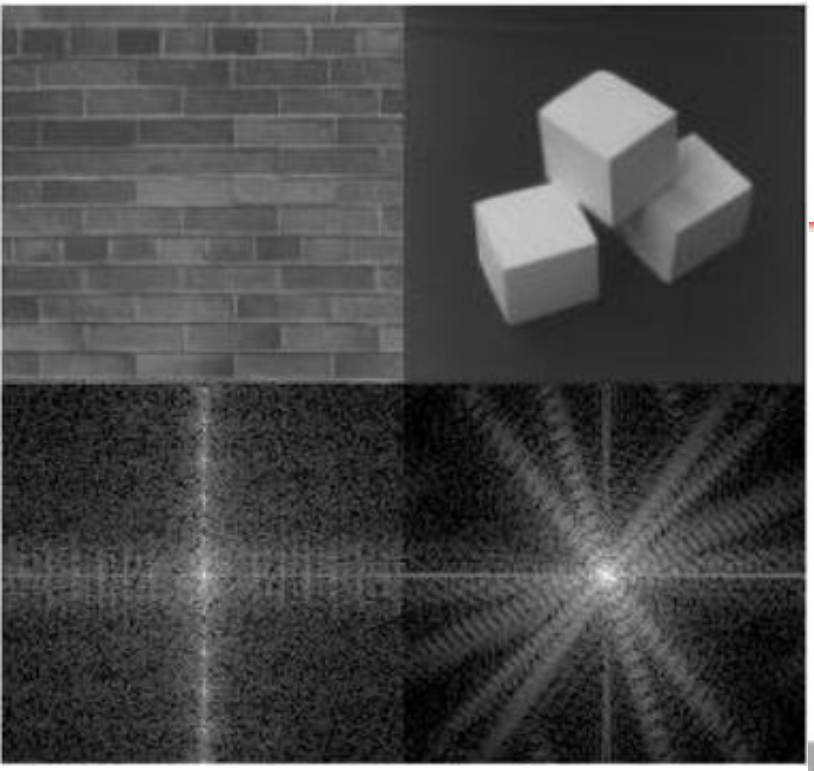
FIGURE 4.27 Left column: Spatial convolution computed with Eq. (3-44), using the approach discussed in Section 3.4. Right column: Circular convolution. The solid line in (j) is the result we would obtain using the DFT, or, equivalently, Eq. (4-48). This erroneous result can be remedied by using zero padding.

Consider two functions f(x) and h(x) composed of A and B samples, respectively. If we append zeros to both functions so that they have the same length, denoted by P, then wraparound is avoided by choosing $P \geq A+B-1$

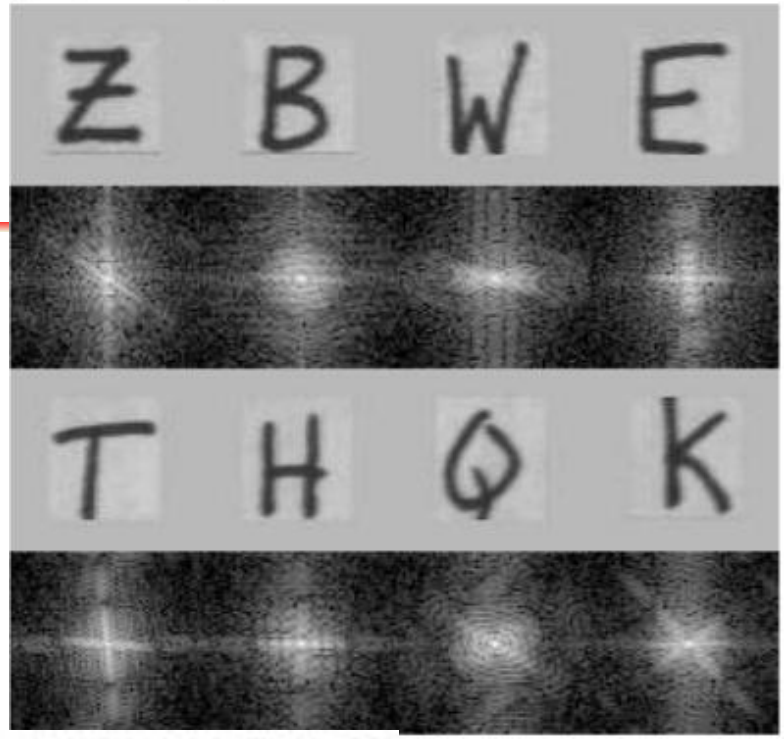
2D DFT Summary

TABLE 4.3 Summary of DFT definitions and corresponding expressions.

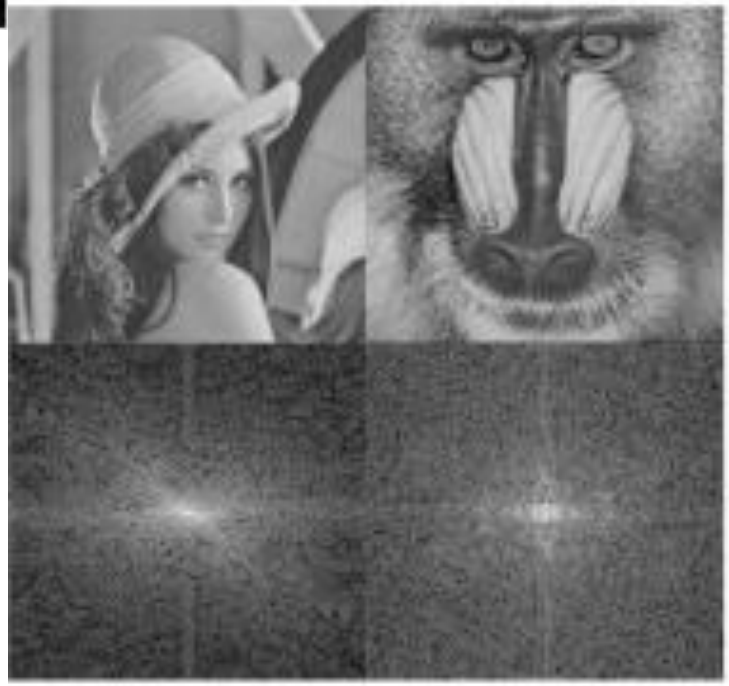
Name	Expression(s)
1) Discrete Fourier transform (DFT) of $f(x,y)$	$F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$
2) Inverse discrete Fourier transform (IDFT) of $F(u,v)$	$f(x,y) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(ux/M+vy/N)}$
3) Spectrum	$\lvert F(u,v)\rvert = \left[R^2(u,v) + I^2(u,v)\right]^{1/2}$ $R = \text{Real}(F);\ I = \text{Imag}(F)$
4) Phase angle	$\phi(u,v) = \tan^{-1}\left[\frac{I(u,v)}{R(u,v)}\right]$
5) Polar representation	$F(u,v) = \lvert F(u,v)\rvert e^{j\phi(u,v)}$
6) Power spectrum	$P(u,v) = \lvert F(u,v)\rvert^2$
7) Average value	$\bar{f} = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y) = \frac{1}{MN}F(0,0)$
8) Periodicity (k_1 and k_2 are integers)	$F(u,v) = F(u+k_1M,v) = F(u,v+k_2N)$ $= F(u+k_1,v+k_2N)$ $f(x,y) = f(x+k_1M,y) = f(x,y+k_2N)$ $= f(x+k_1M,y+k_2N)$
9) Convolution	$(f \star h)(x,y) = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f(m,n)h(x-m,y-n)$
10) Correlation	$(f \star h)(x,y) = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f^*(m,n)h(x+m,y+n)$
11) Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.
12) Obtaining the IDFT using a DFT algorithm	$MNf^*(x,y) = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F^*(u,v)e^{-j2\pi(ux/M+vy/N)}$ This equation indicates that inputting $F^*(u,v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MNf^*(x,y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.



明亮线与图像中对应的轮廓线是垂直的。若图像中有圆形区域，则频谱也呈圆形分布

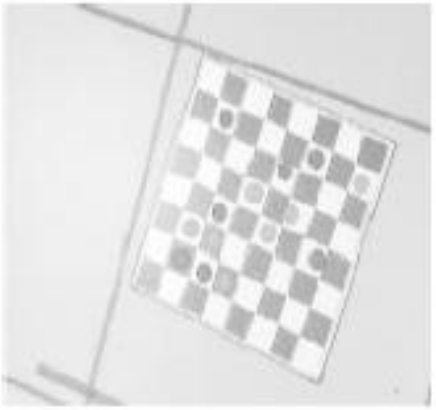


频谱中明亮线反映出图像的灰度级变化

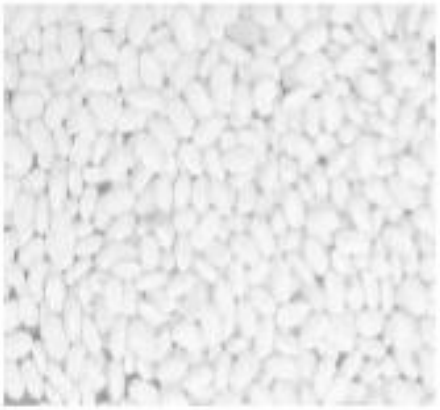


原图没有明显结构，观察原图细节：人物图帽子和头发的边线—频谱中的斜线

1. 下面有三幅内容分别为“棋盘”、“花生”和“工具”的图像以及它们的 DFT 幅度谱(分别标记为 a, b, c)，请指出频谱 (a), (b), (c) 与三幅图像之间的对应关系，并解释为什么。



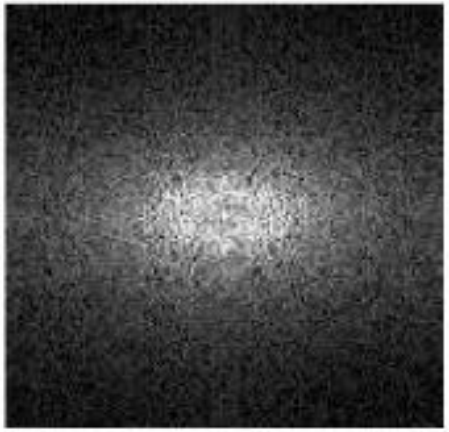
棋盘



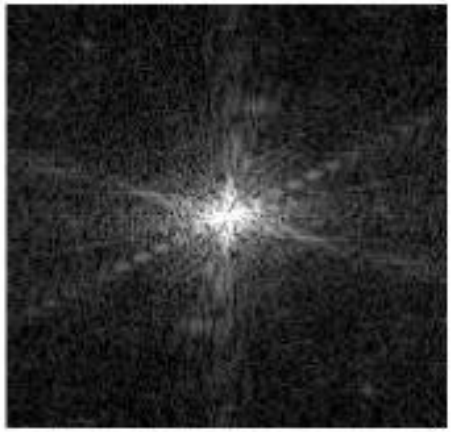
花生



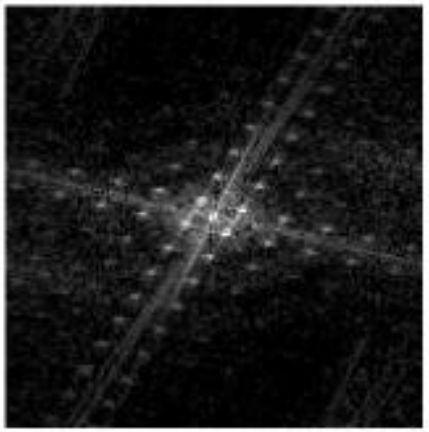
工具



(a)



(b)



(c)

……



Check Your Understanding

Predict the spectrum of horizontal and vertical stripes.

Explain why aliasing prevents unique reconstruction.

Explain what zero padding does and does not improve.

Describe how translation changes magnitude and phase.

Choose a padding size for linear convolution.

Summary

- 加深频谱的理解
  - 对图像进行DFT得到的频谱图，就是图像梯度的分布图
  - 频谱图上各点与图像各点不存在一一对应关系
  - 频谱图上明暗不一的亮点，表征是图像上某一点与邻域点差异的强弱（梯度的大小），也称为该点(u,v)的频率的大小
  - 图像中的低频部分指低梯度的点；高频相反
    - 若频谱图中暗点多，表示实际图像比较柔和；
    - 若频谱图中亮点多，则实际图像是尖锐的，边界分明且边界两边像素灰度差异大
  - 频率谱仅包含灰度信息，没有形状信息
  - 相位谱含有图像中的位置信息